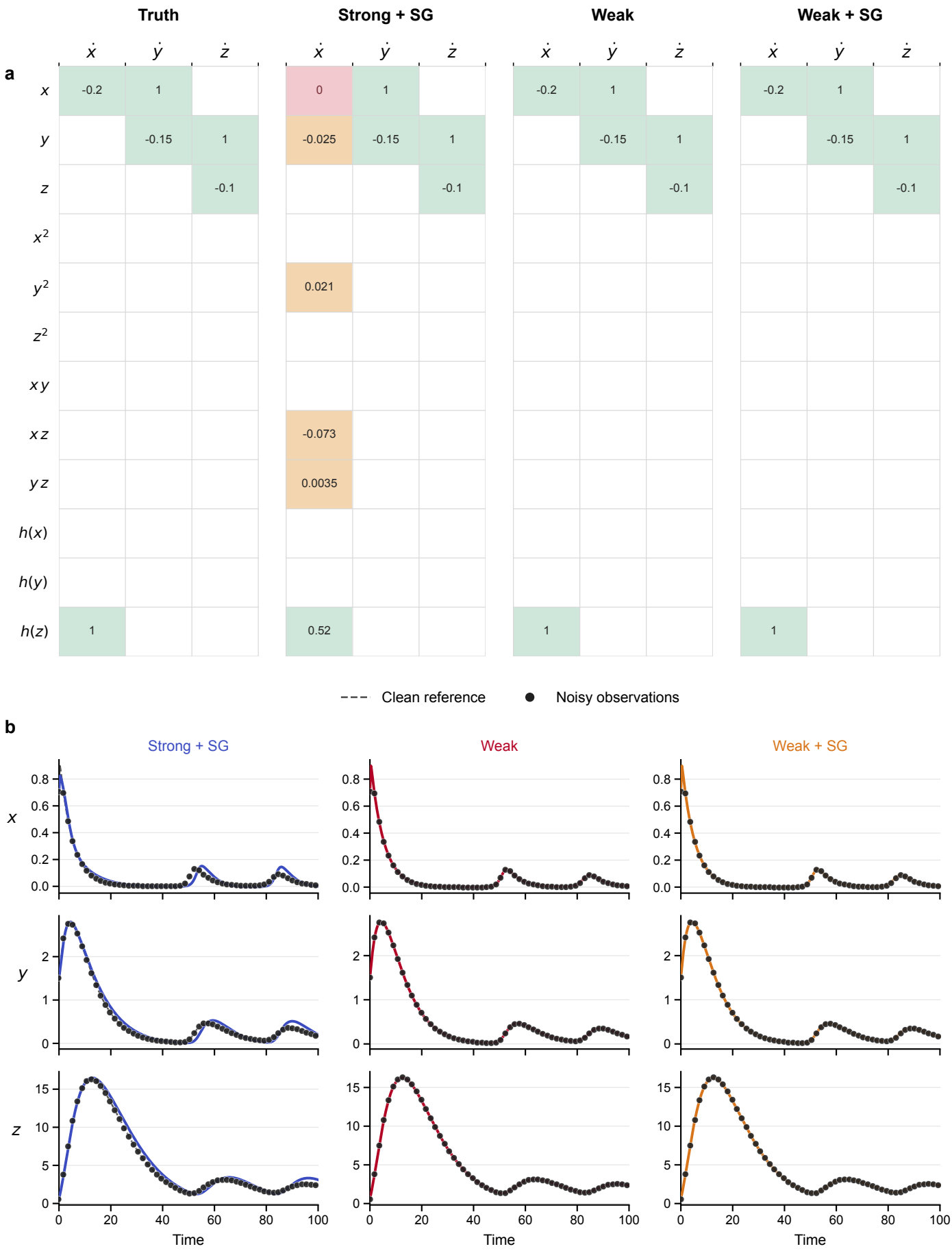
## Figure S1 | 0% noise, seed 11

Selected coefficient matrices and first held-out trajectory

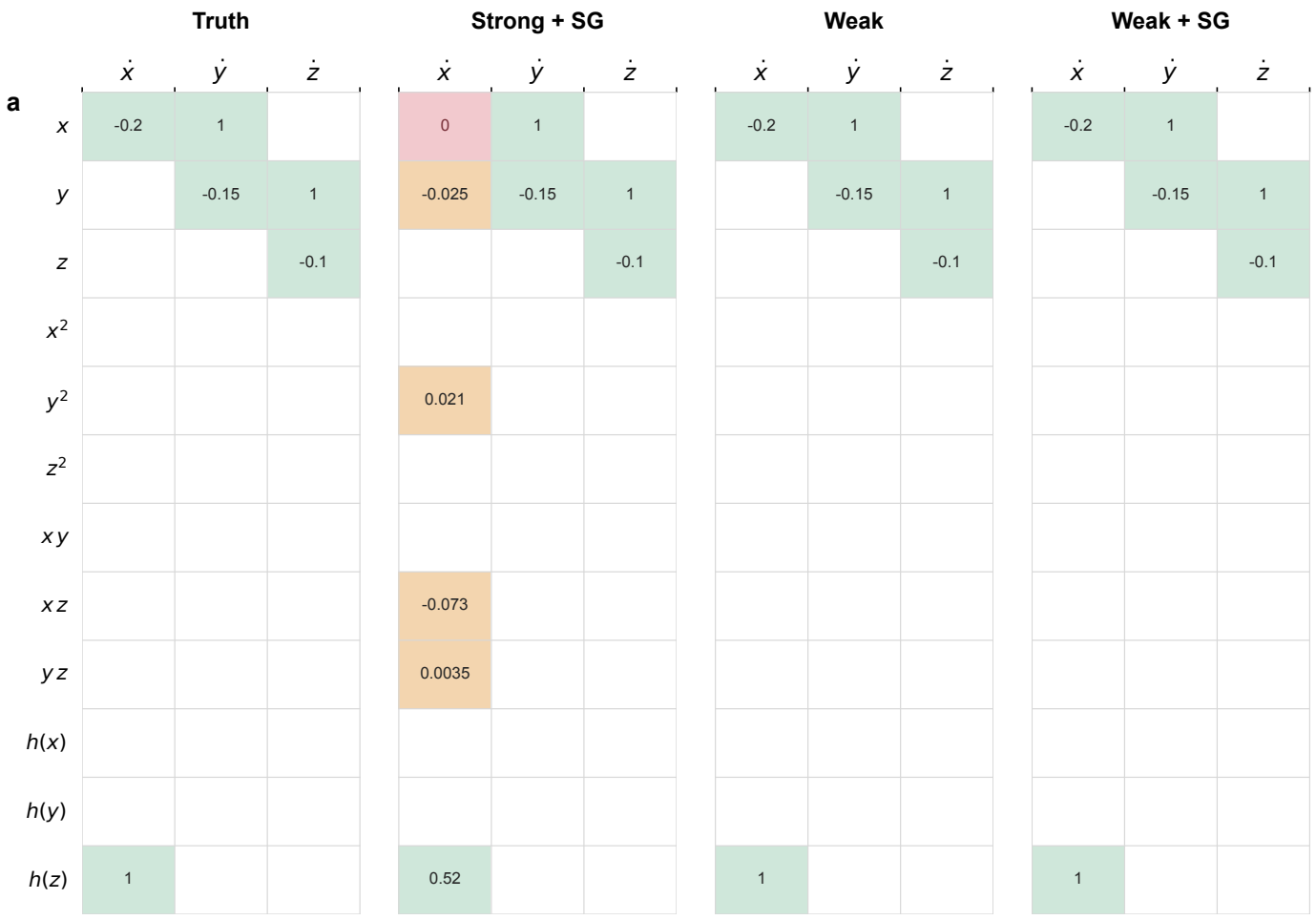


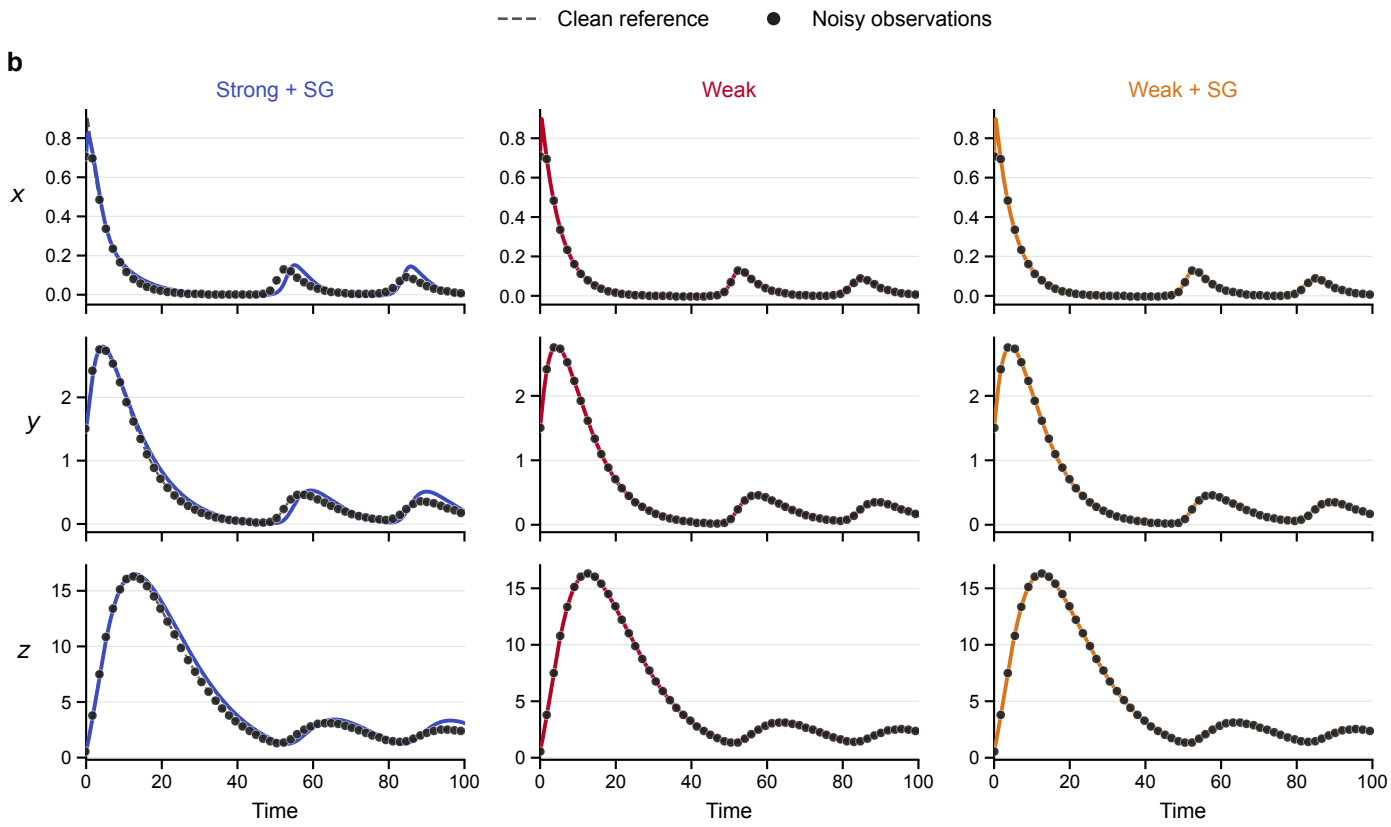

Green = true positive; orange = false positive; rose = missed term. Numbers show coefficients.

Dots show noisy test observations; colored lines show selected-model rollouts.

## Figure S1 | 0% noise, seed 22

Selected coefficient matrices and first held-out trajectory



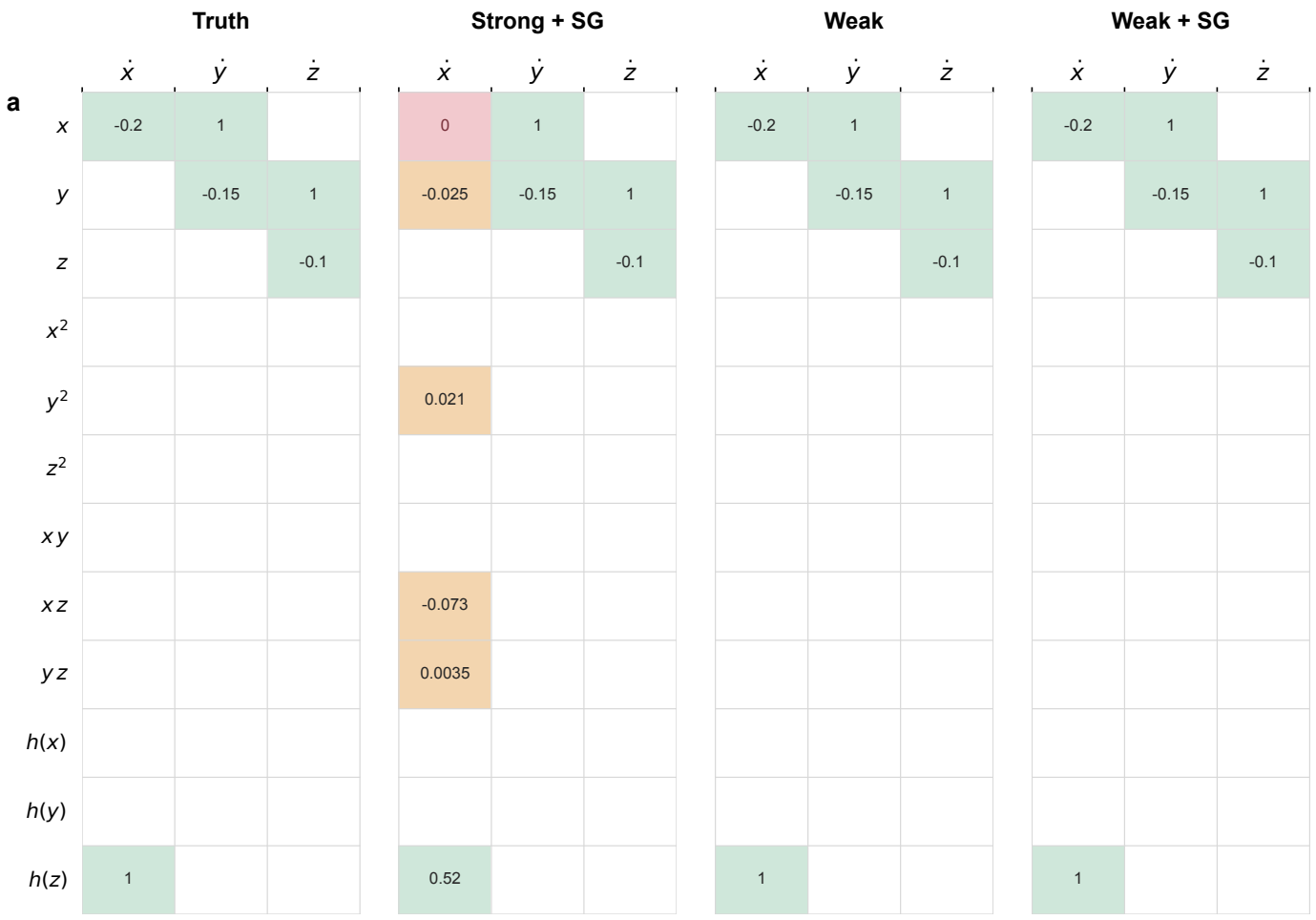



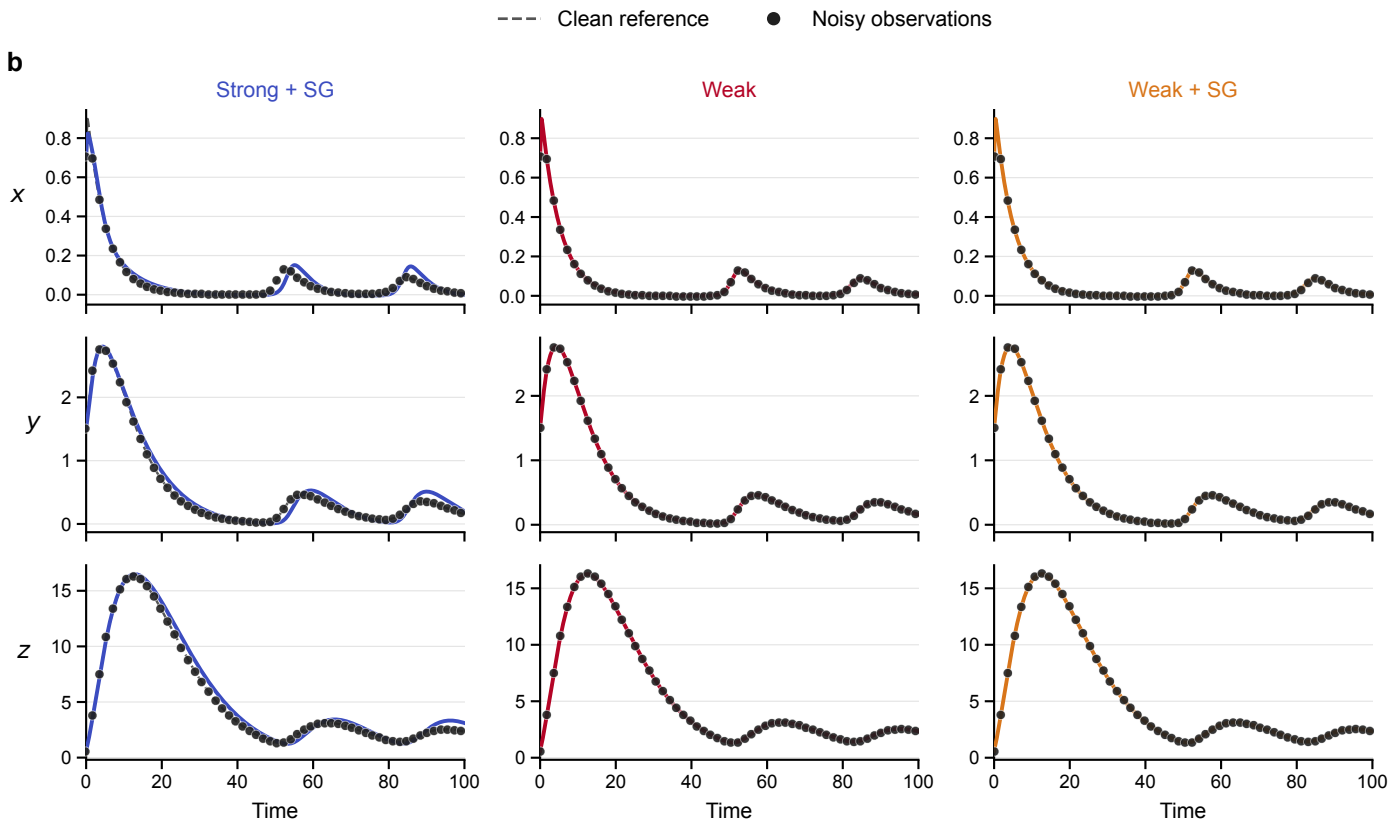

Green = true positive; orange = false positive; rose = missed term. Numbers show coefficients.

Dots show noisy test observations; colored lines show selected-model rollouts.

## Figure S1 | 0% noise, seed 33

Selected coefficient matrices and first held-out trajectory



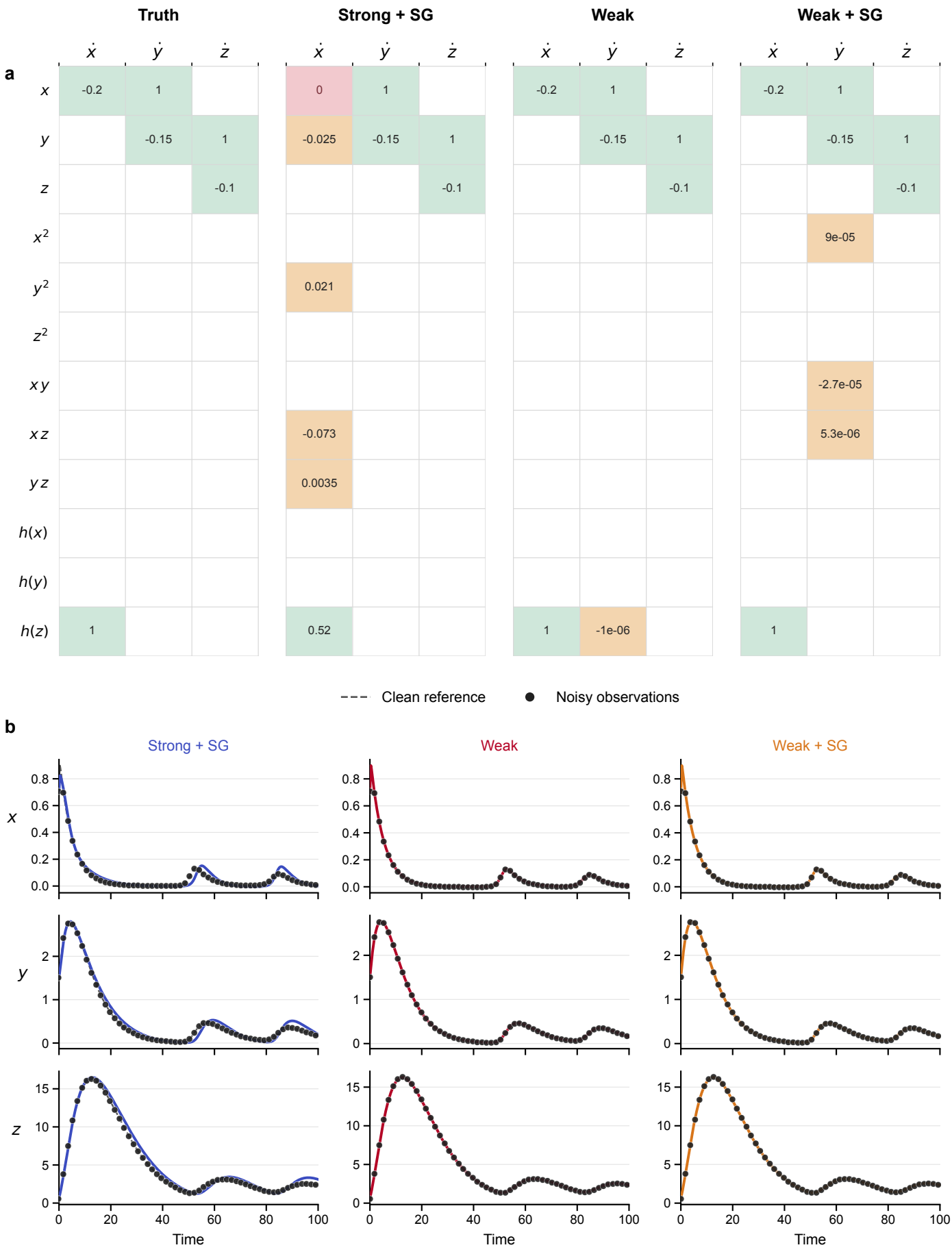




Green = true positive; orange = false positive; rose = missed term. Numbers show coefficients.

Dots show noisy test observations; colored lines show selected-model rollouts.

## Figure S1 | 0% noise, seed 44

Selected coefficient matrices and first held-out trajectory

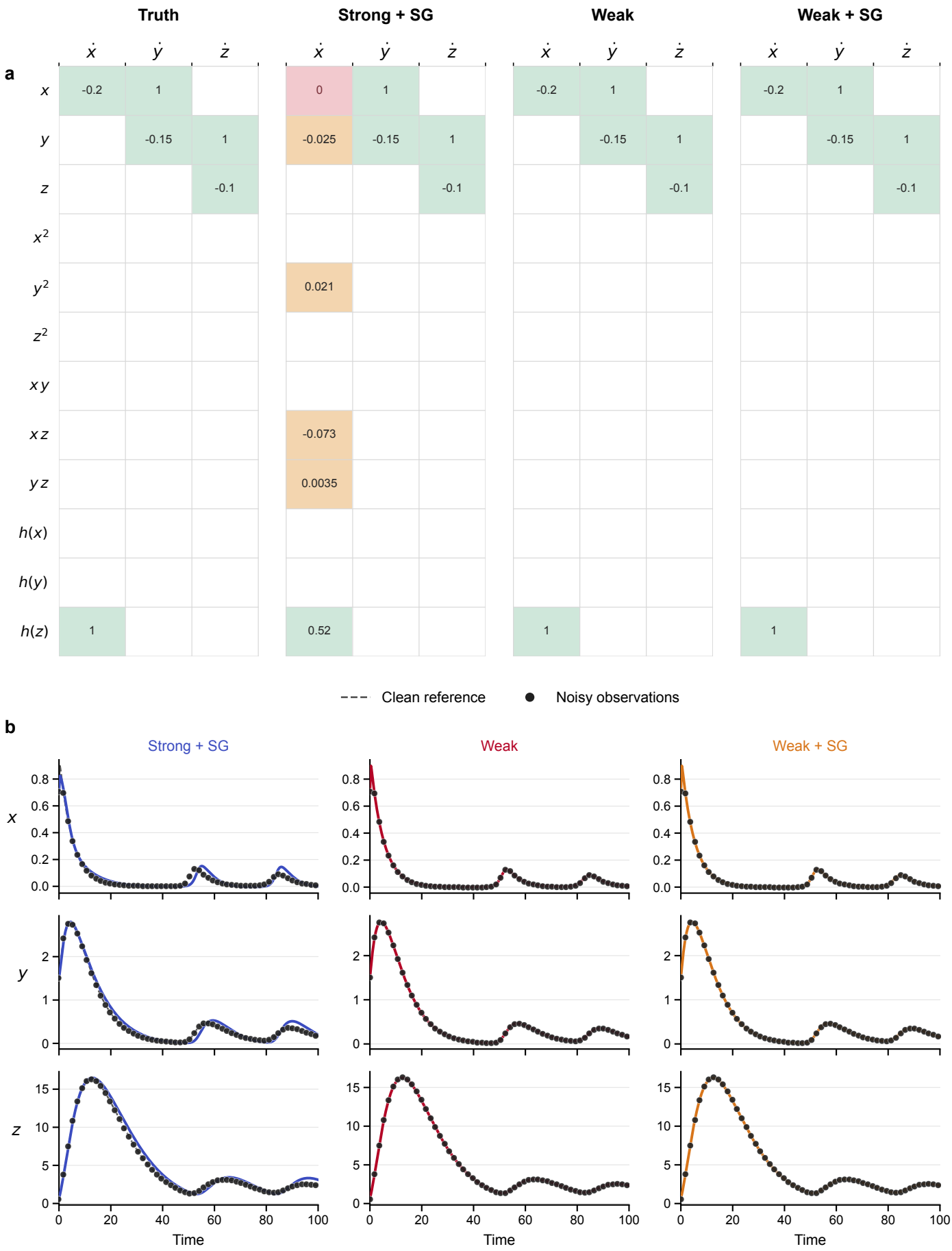



Green = true positive; orange = false positive; rose = missed term. Numbers show coefficients.

Dots show noisy test observations; colored lines show selected-model rollouts.

## Figure S1 | 0% noise, seed 55

Selected coefficient matrices and first held-out trajectory

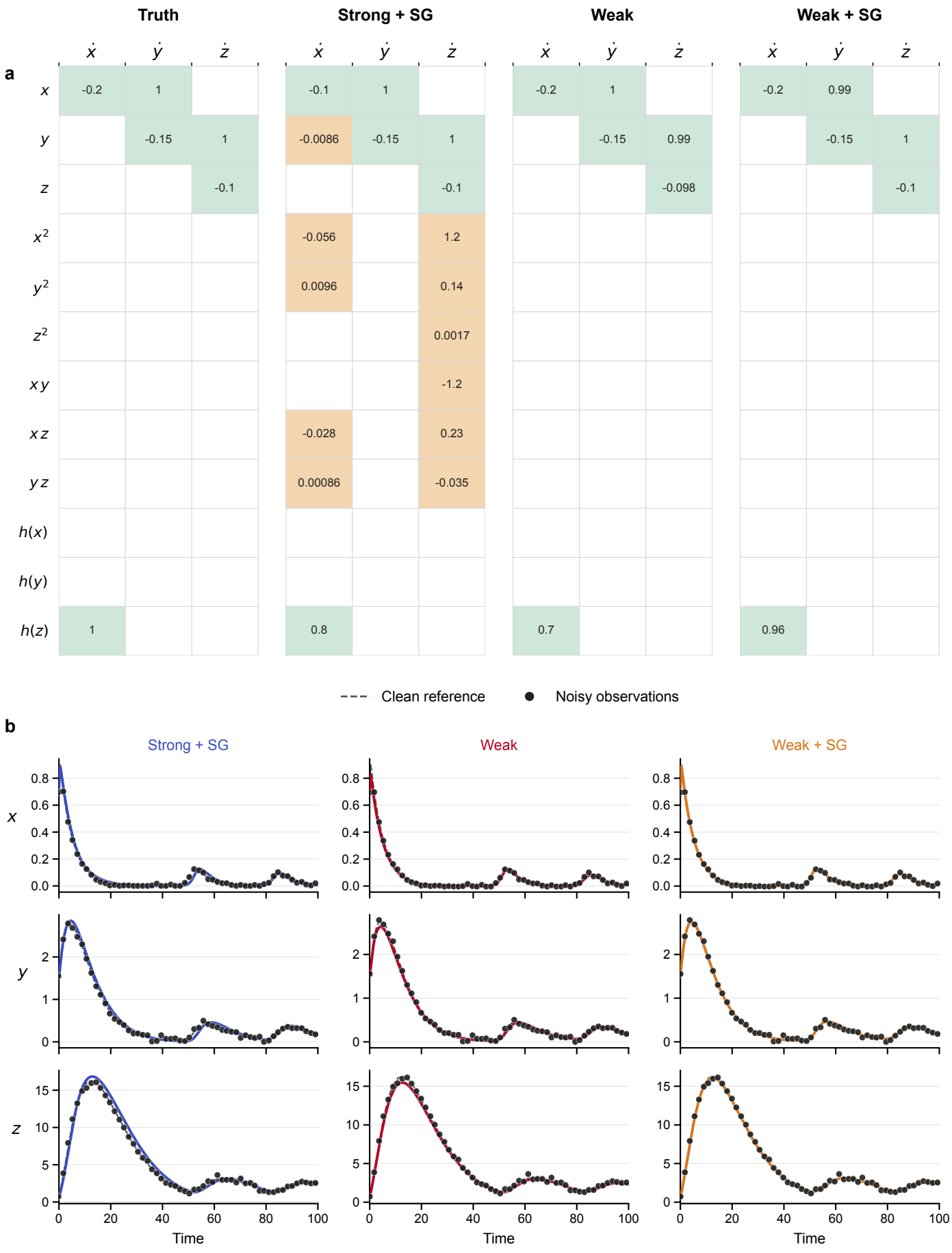



Green = true positive; orange = false positive; rose = missed term. Numbers show coefficients.

Dots show noisy test observations; colored lines show selected-model rollouts.

## Figure S1 | 5% noise, seed 11

Selected coefficient matrices and first held-out trajectory




Green = true positive; orange = false positive; rose = missed term. Numbers show coefficients.

Dots show noisy test observations; colored lines show selected-model rollouts.

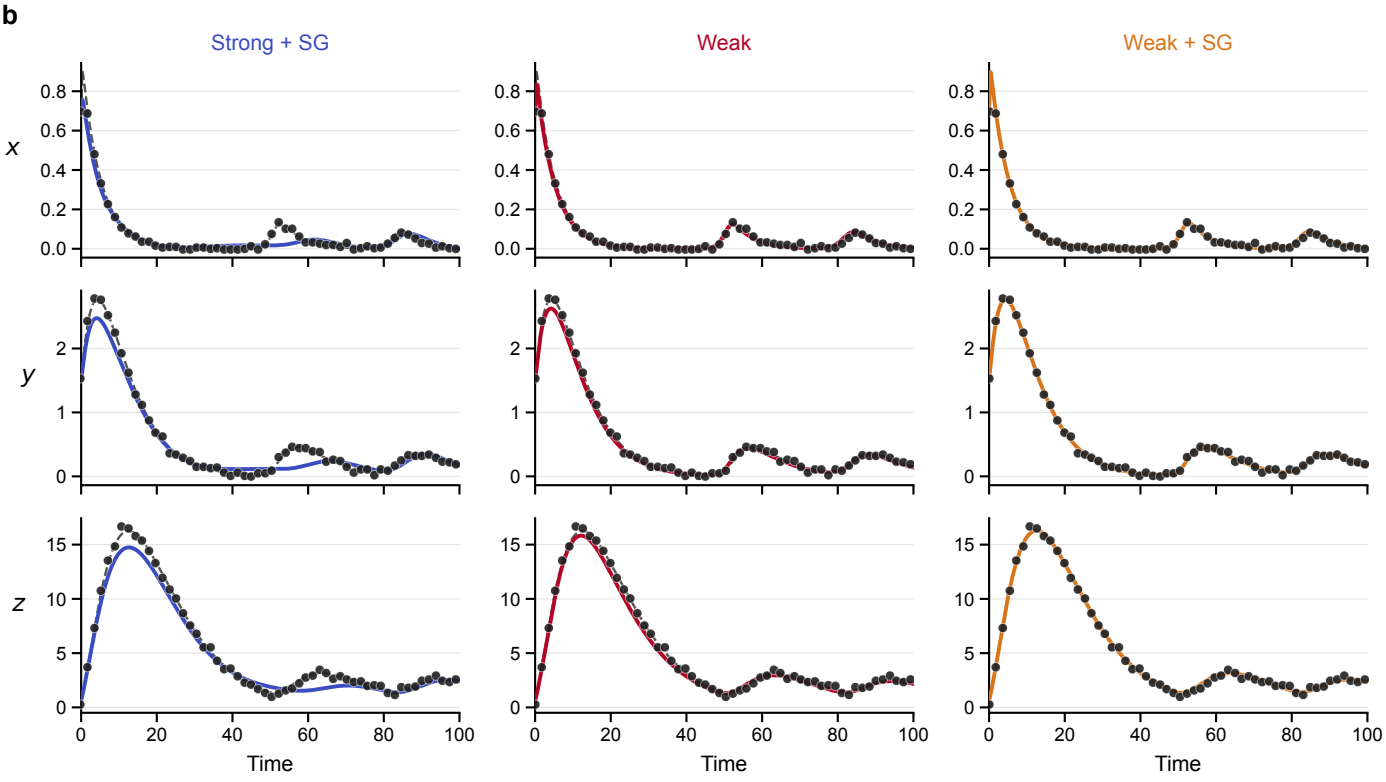
## Figure S1 | 5% noise, seed 22

Selected coefficient matrices and first held-out trajectory

**a**

| | Truth | | | Strong + SG | | | Weak | | | Weak + SG | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ |
| $x$ | -0.2 | 1 | | 0.14 | 1 | | -0.2 | 1 | | -0.2 | 1 | |
| $y$ | | -0.15 | 1 | -0.053 | -0.15 | 1 | | -0.15 | 0.93 | | -0.15 | 1 |
| $z$ | | | -0.1 | 0.0023 | | -0.1 | | | -0.094 | | | -0.1 |
| $x^2$ | | | | -0.29 | | | | | | | | |
| $y^2$ | | | | 0.021 | | | | | | | | |
| $z^2$ | | | | -0.00017 | | | | | -0.0013 | | | |
| $xy$ | | | | | | | | | | | | |
| $xz$ | | | | -0.055 | | | | | | | | |
| $yz$ | | | | 0.0037 | | | | | 0.014 | | | |
| $h(x)$ | | | | | | | | | | | | |
| $h(y)$ | | | | | | | | | | | | |
| $h(z)$ | 1 | | | 0.38 | | | 0.72 | | | 1 | | |

**b**

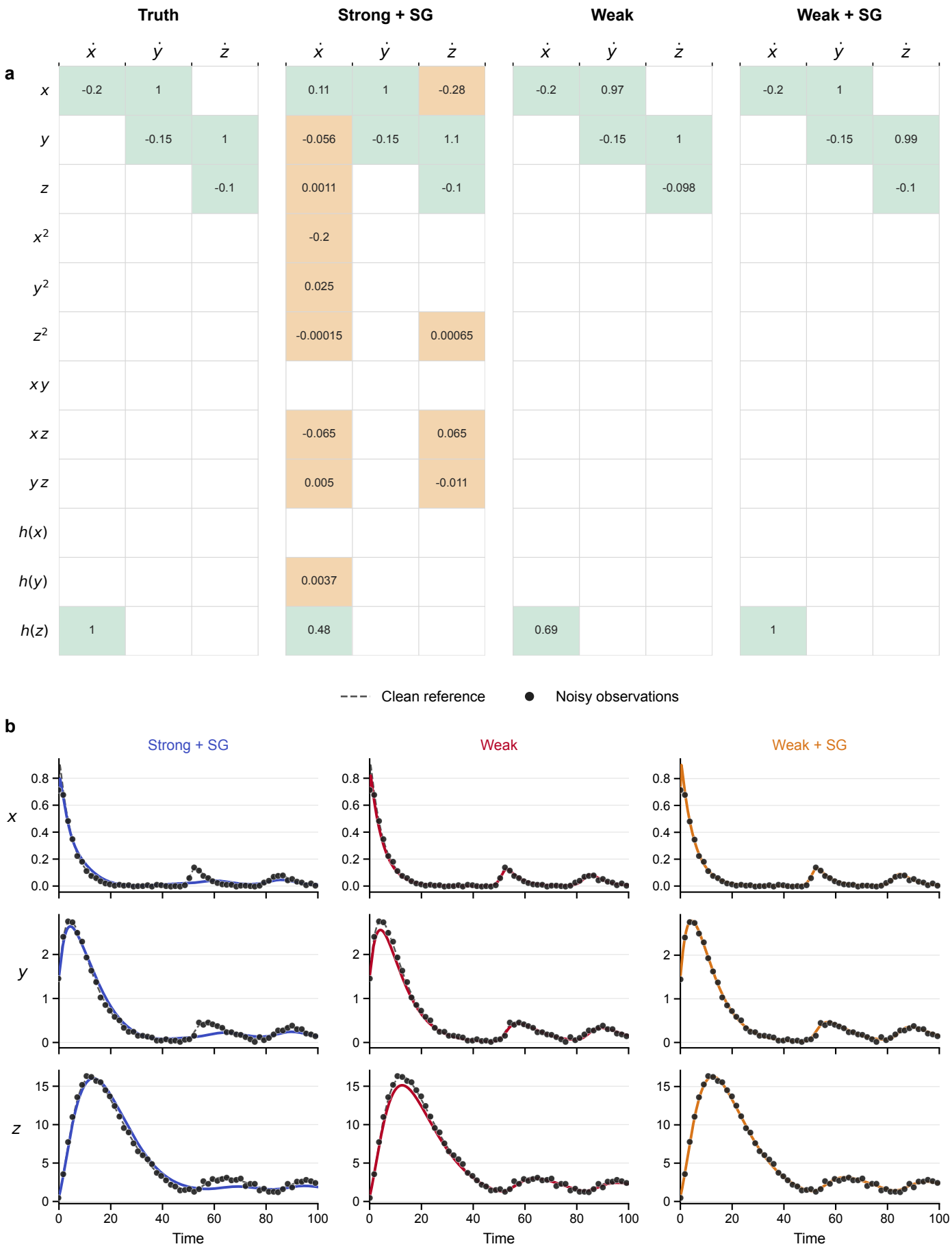



Green = true positive; orange = false positive; rose = missed term. Numbers show coefficients.

Dots show noisy test observations; colored lines show selected-model rollouts.

## Figure S1 | 5% noise, seed 33

Selected coefficient matrices and first held-out trajectory

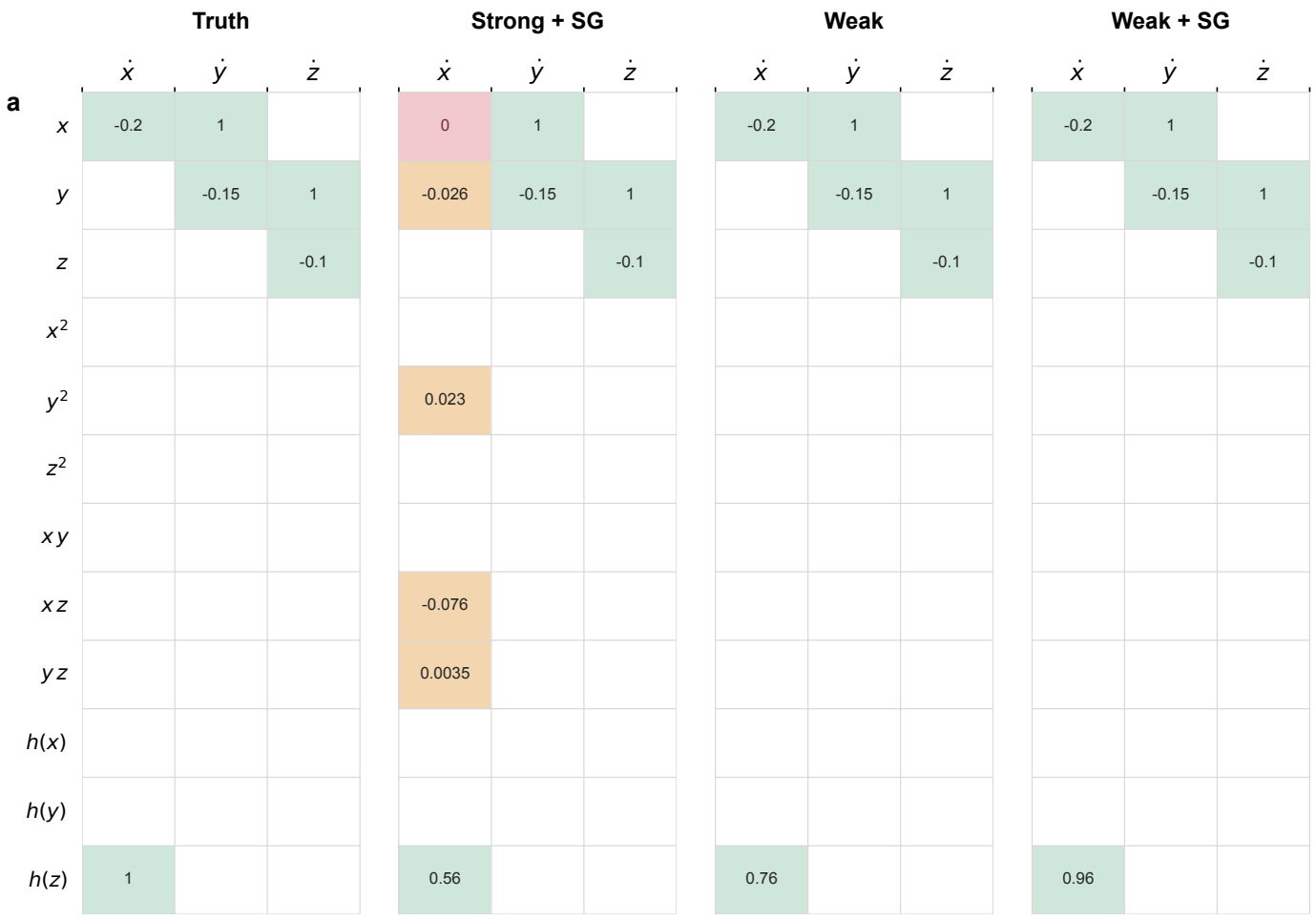


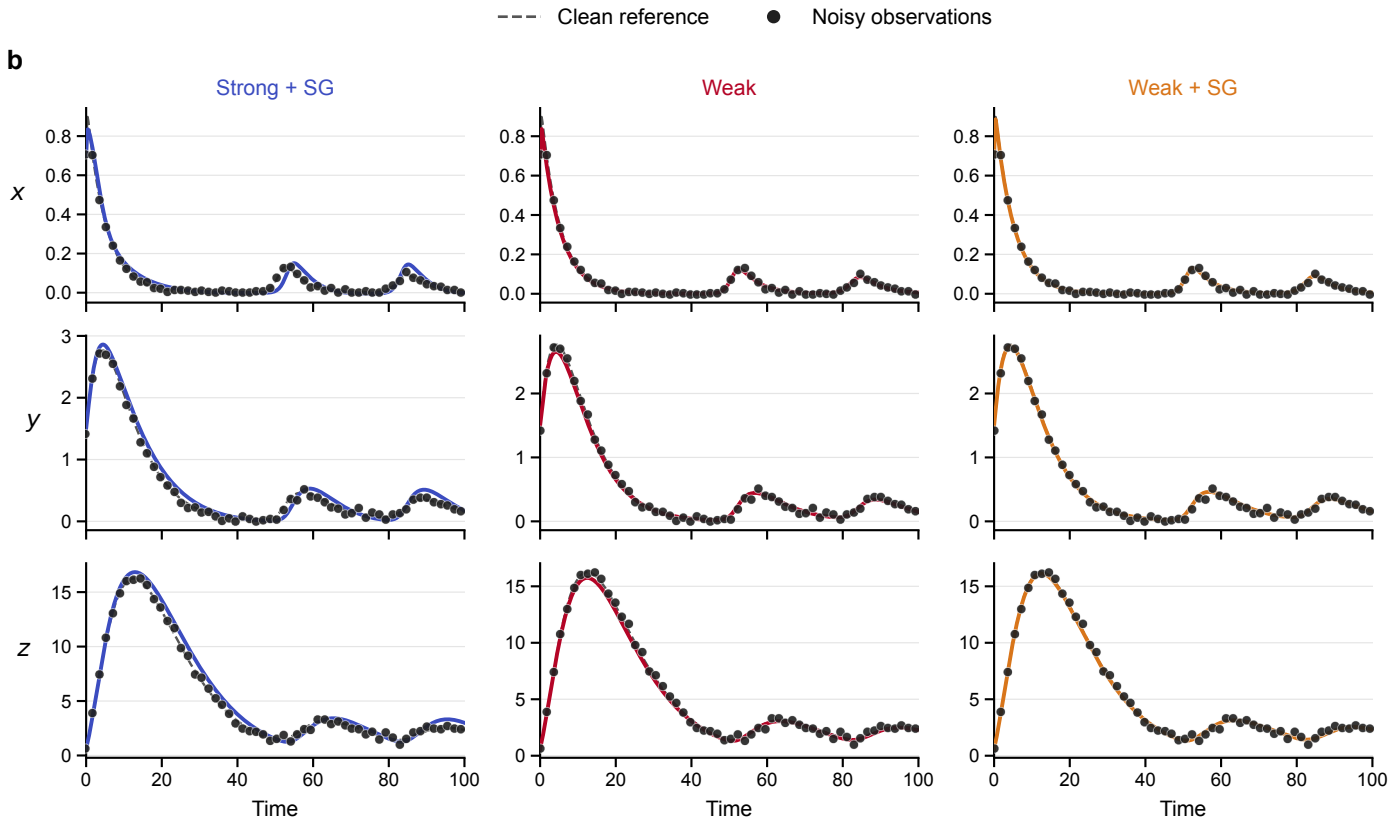

Green = true positive; orange = false positive; rose = missed term. Numbers show coefficients.

Dots show noisy test observations; colored lines show selected-model rollouts.

## Figure S1 | 5% noise, seed 44

Selected coefficient matrices and first held-out trajectory




Green = true positive; orange = false positive; rose = missed term. Numbers show coefficients.

Dots show noisy test observations; colored lines show selected-model rollouts.

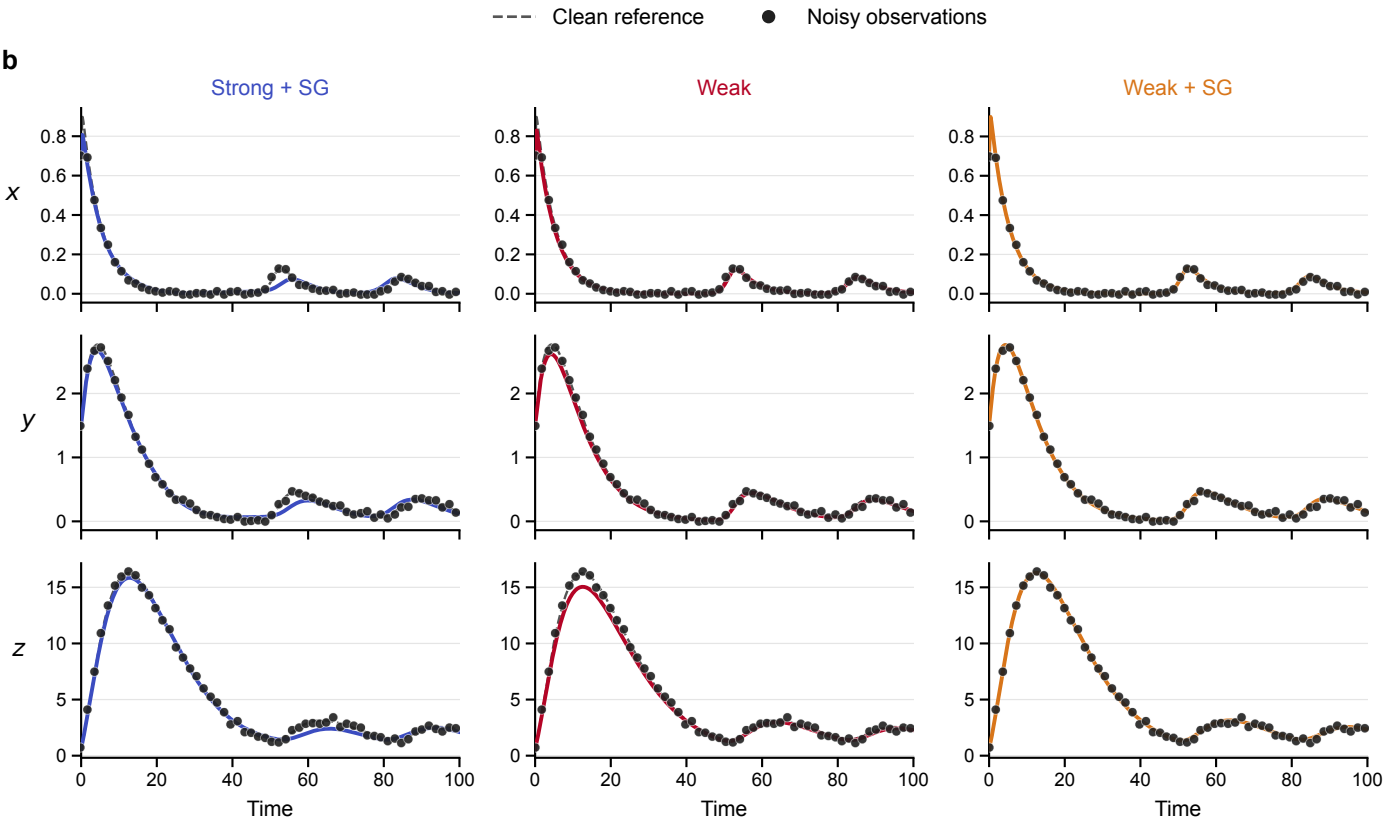
## Figure S1 | 5% noise, seed 55

Selected coefficient matrices and first held-out trajectory

**a**

| | Truth | | | Strong + SG | | | Weak | | | Weak + SG | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ |
| $x$ | -0.2 | 1 | | 0 | 1.1 | | -0.2 | 1 | | -0.2 | 1 | |
| $y$ | | -0.15 | 1 | -0.03 | -0.16 | 0.99 | | -0.15 | 0.96 | | -0.15 | 1 |
| $z$ | | | -0.1 | 0.0014 | | -0.1 | | | -0.097 | | | -0.1 |
| $x^2$ | | | | -0.21 | -0.13 | 0.88 | | | | | | |
| $y^2$ | | | | 0.0099 | -0.008 | 0.04 | | | | | | |
| $z^2$ | | | | -0.00014 | -0.00014 | | | | | | | |
| $xy$ | | | | 0.043 | 0.1 | -0.4 | | | | | | |
| $xz$ | | | | -0.044 | -0.04 | | | | | | | |
| $yz$ | | | | 0.003 | 0.0043 | | | | | | | |
| $h(x)$ | | | | | | | | | | | | |
| $h(y)$ | | | | | | | | | | | | |
| $h(z)$ | 1 | | | 0.62 | | | 0.71 | | | 1 | | |

**b**

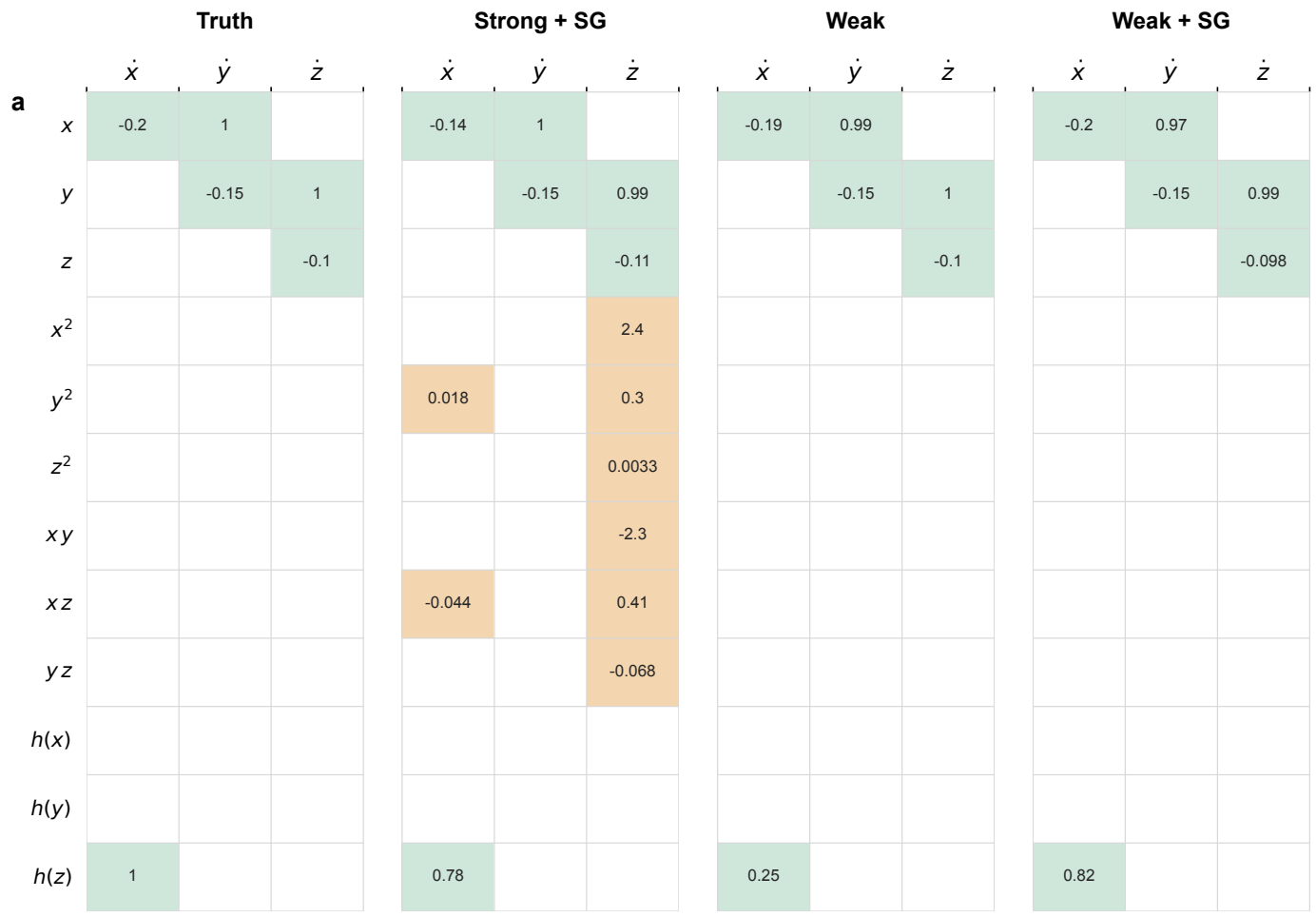


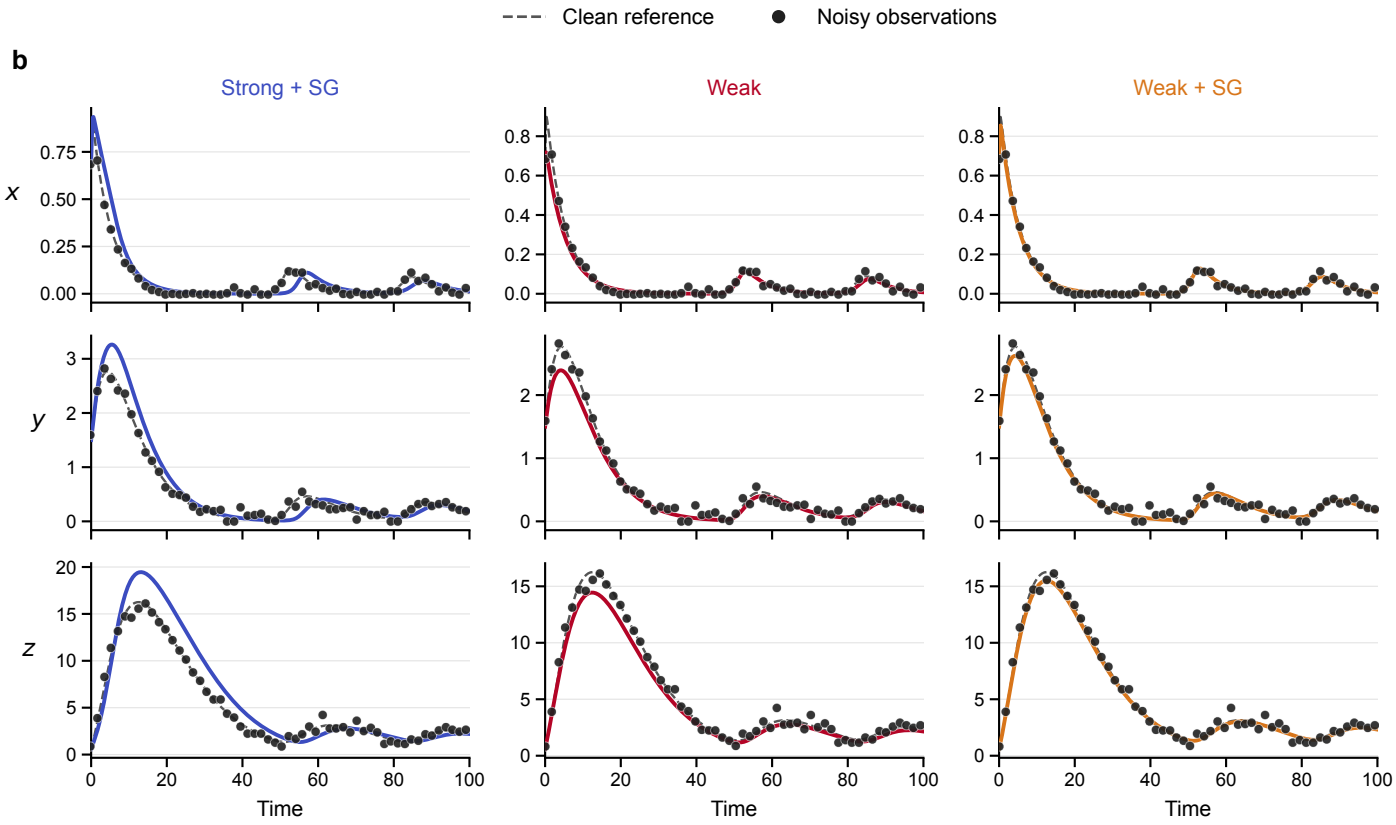

Green = true positive; orange = false positive; rose = missed term. Numbers show coefficients.

Dots show noisy test observations; colored lines show selected-model rollouts.

## Figure S1 | 10% noise, seed 11

Selected coefficient matrices and first held-out trajectory




Green = true positive; orange = false positive; rose = missed term. Numbers show coefficients.

Dots show noisy test observations; colored lines show selected-model rollouts.

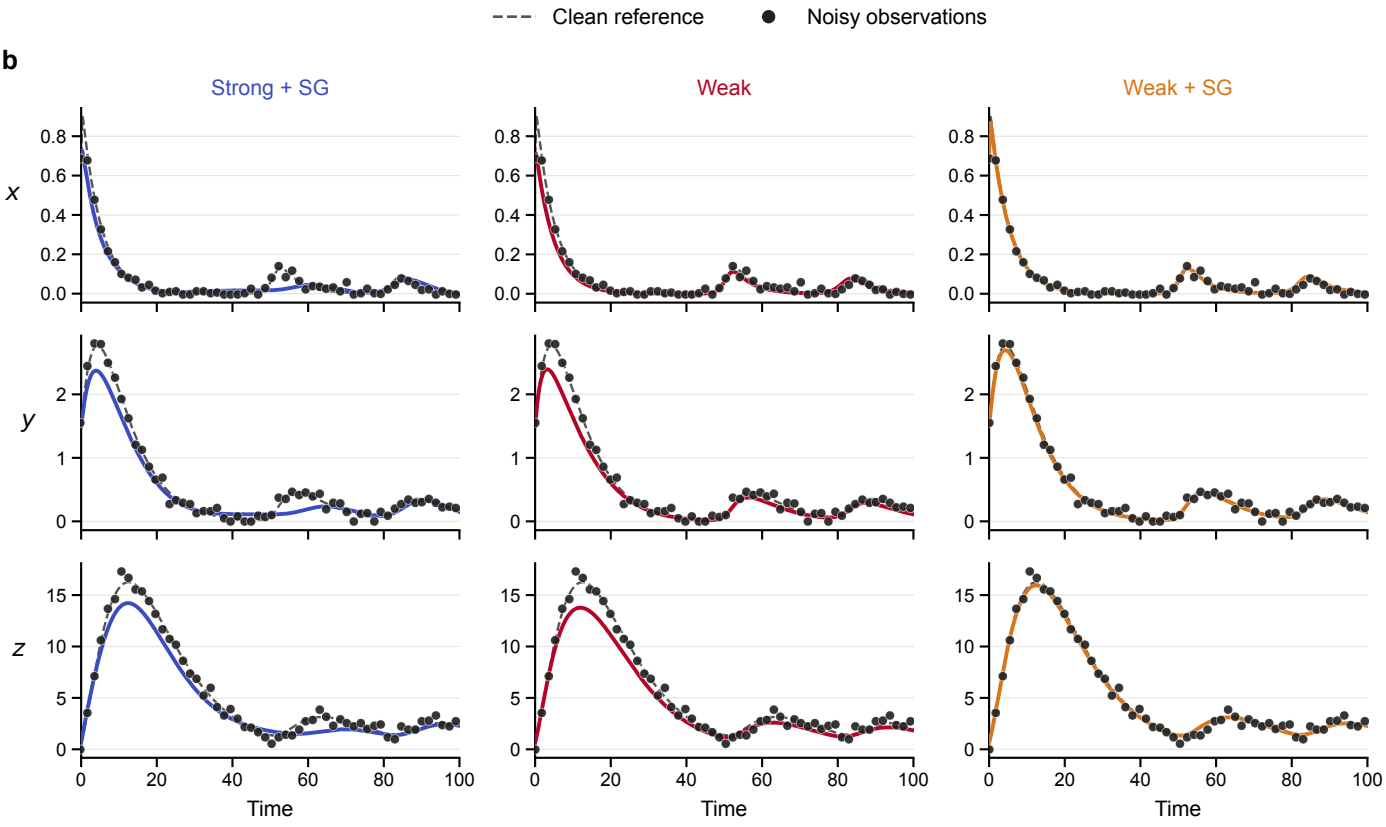
## Figure S1 | 10% noise, seed 22

Selected coefficient matrices and first held-out trajectory

**a**

| | Truth | | | Strong + SG | | | Weak | | | Weak + SG | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ |
| $x$ | -0.2 | 1 | | 0.16 | 1 | 0.64 | -0.21 | 1 | | -0.2 | 1 | |
| $y$ | | -0.15 | 1 | -0.055 | -0.15 | 0.88 | | -0.15 | 1 | | -0.15 | 1 |
| $z$ | | | -0.1 | 0.0023 | | -0.092 | | | -0.1 | | | -0.1 |
| $x^2$ | | | | -0.3 | | | | 0.99 | | | | |
| $y^2$ | | | | 0.025 | | | | 0.064 | | | | |
| $z^2$ | | | | -0.00016 | | -0.0025 | | | | | | |
| $xy$ | | | | | | 0.49 | | -0.36 | | | | |
| $xz$ | | | | -0.064 | | -0.44 | | -0.084 | | | | |
| $yz$ | | | | 0.0036 | | 0.05 | | | | | | |
| $h(x)$ | | | | | | | | | | | | |
| $h(y)$ | | | | | | | | | | | | |
| $h(z)$ | 1 | | | 0.28 | | | 0.22 | | | 0.88 | | |

**b**

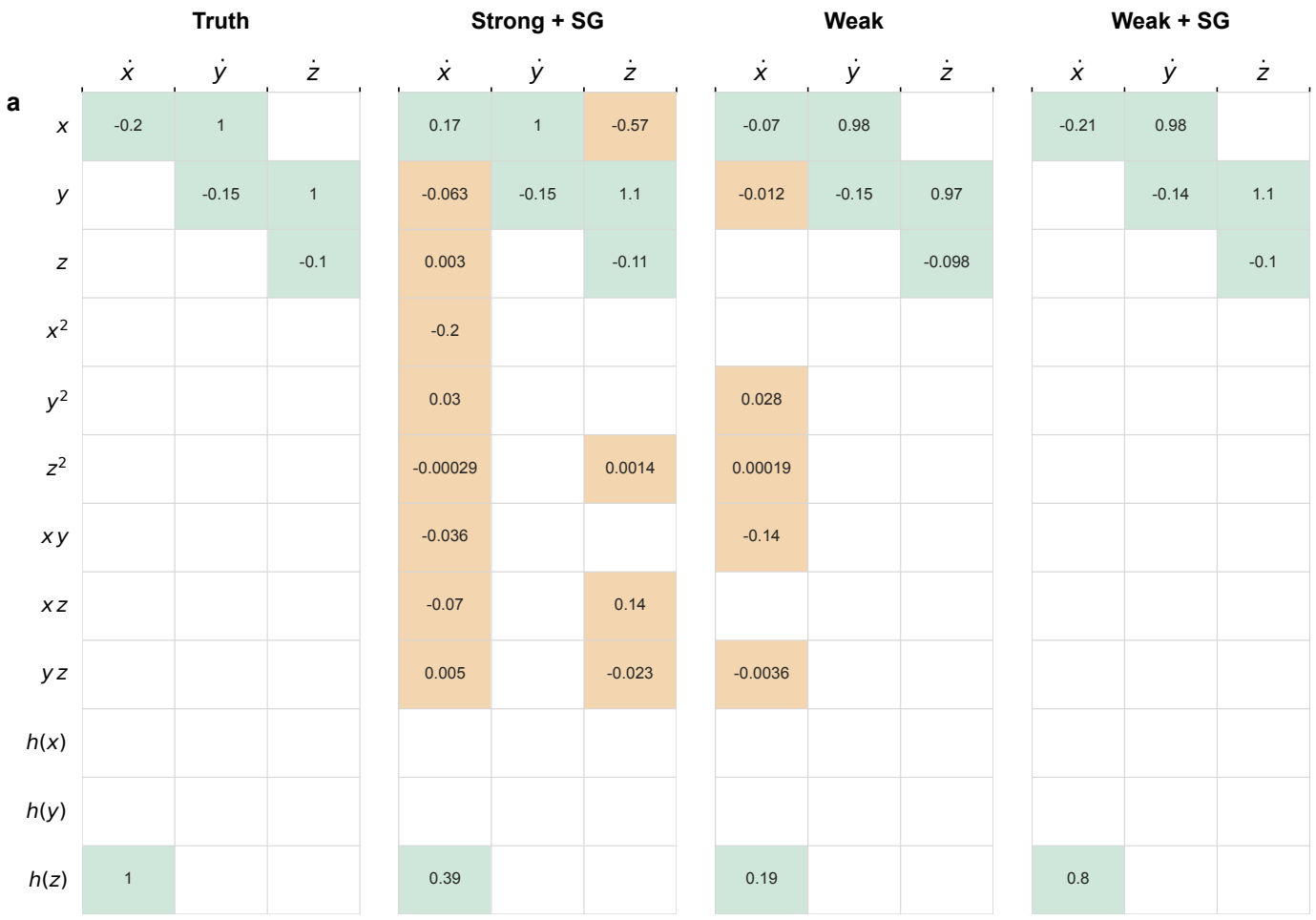


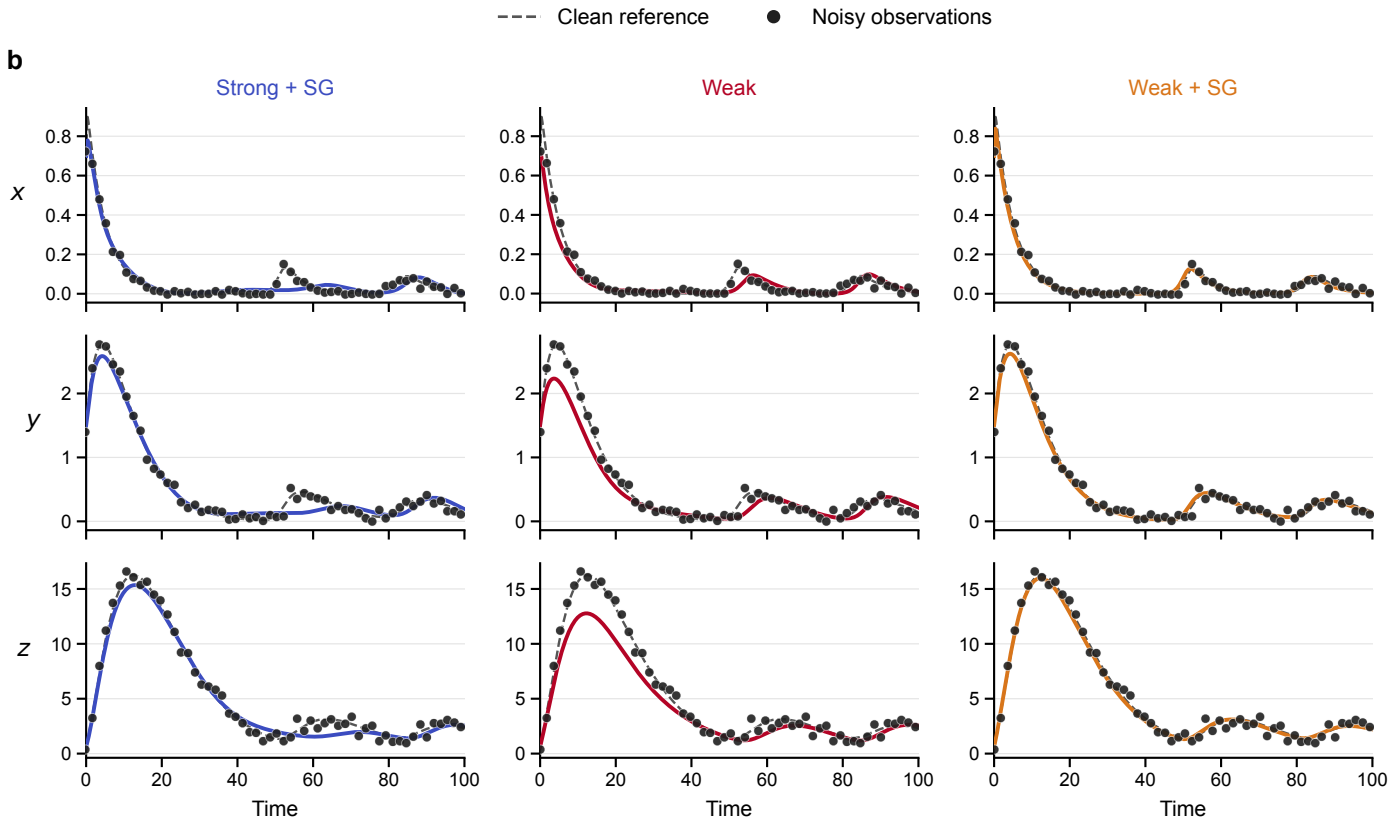

Green = true positive; orange = false positive; rose = missed term. Numbers show coefficients.

Dots show noisy test observations; colored lines show selected-model rollouts.

## Figure S1 | 10% noise, seed 33

Selected coefficient matrices and first held-out trajectory




Green = true positive; orange = false positive; rose = missed term. Numbers show coefficients.

Dots show noisy test observations; colored lines show selected-model rollouts.

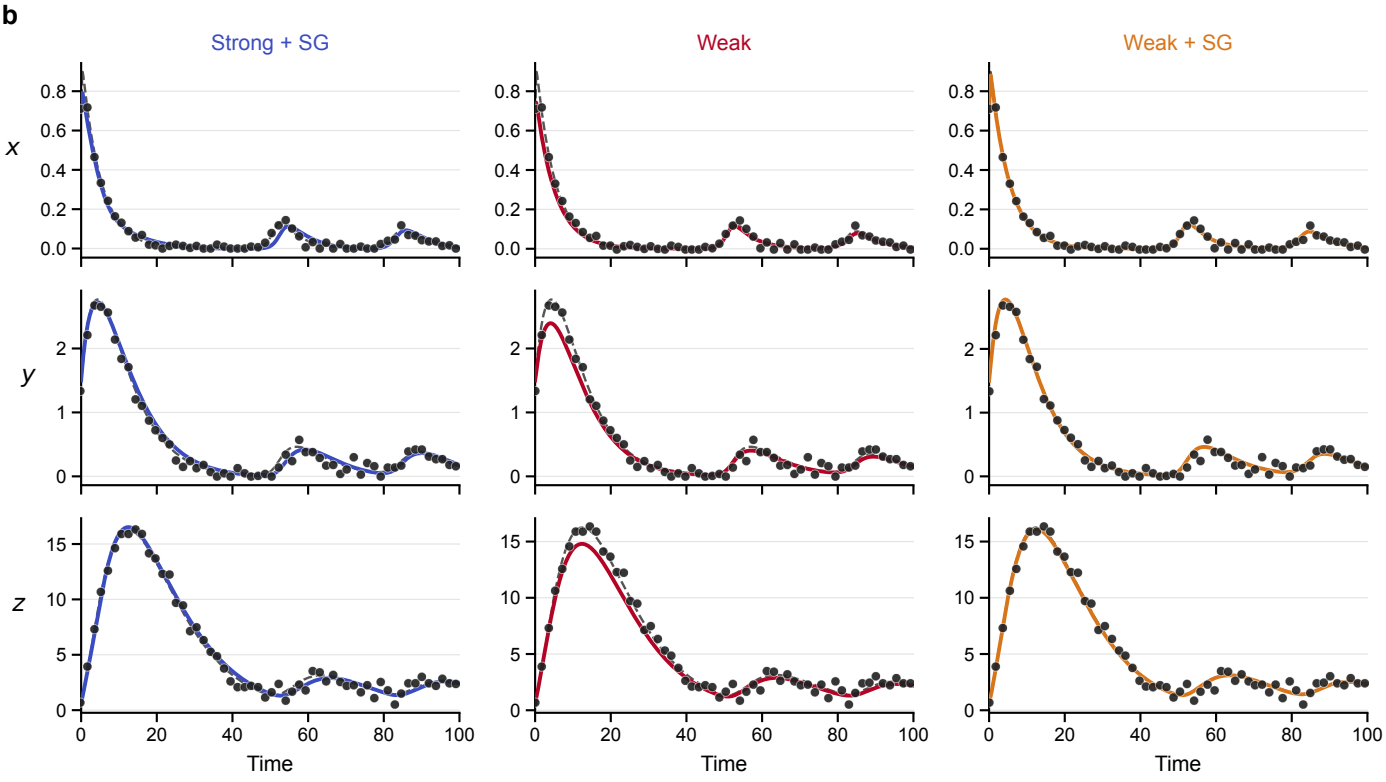
## Figure S1 | 10% noise, seed 44

Selected coefficient matrices and first held-out trajectory

**a**

| | Truth | | | Strong + SG | | | Weak | | | Weak + SG | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ |
| $x$ | -0.2 | 1 | | -0.048 | 0.96 | | -0.19 | 0.96 | | -0.2 | 1 | |
| $y$ | | -0.15 | 1 | -0.013 | -0.15 | 0.96 | | -0.15 | 1.1 | | -0.16 | 0.98 |
| $z$ | | | -0.1 | | | -0.095 | | | -0.1 | | | -0.099 |
| $x^2$ | | | | -0.19 | 0.32 | 3.2 | | | | | | |
| $y^2$ | | | | 0.0028 | 0.028 | 0.26 | | | | | | |
| $z^2$ | | | | | | | | | | | | |
| $xy$ | | | | 0.05 | -0.1 | -2 | | | | | | |
| $xz$ | | | | -0.034 | -0.029 | 0.085 | | | | | | |
| $yz$ | | | | 0.0019 | | -0.023 | | | | | | |
| $h(x)$ | | | | | | | | | | | | |
| $h(y)$ | | | | | | | | | | | | |
| $h(z)$ | 1 | | | 0.56 | 0.21 | | 0.36 | | | 0.91 | | |

**b**




Green = true positive; orange = false positive; rose = missed term. Numbers show coefficients.

Dots show noisy test observations; colored lines show selected-model rollouts.

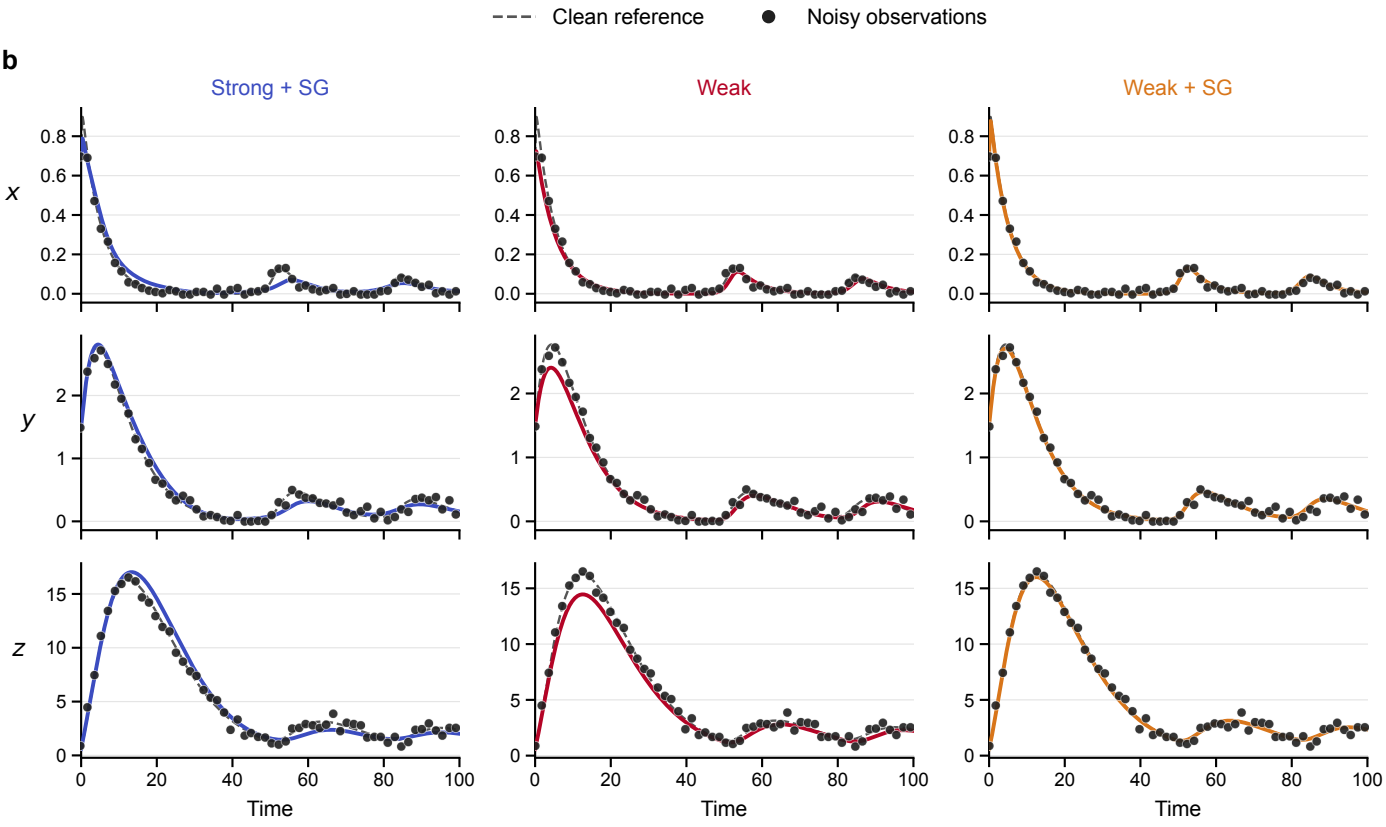
## Figure S1 | 10% noise, seed 55

Selected coefficient matrices and first held-out trajectory

**a**

| | Truth | | | Strong + SG | | | Weak | | | Weak + SG | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ |
| $x$ | -0.2 | 1 | | 0 | 1.1 | | -0.18 | 0.99 | | -0.2 | 0.98 | |
| $y$ | | -0.15 | 1 | -0.028 | -0.16 | 0.98 | | -0.15 | 1 | | -0.15 | 1 |
| $z$ | | | -0.1 | -0.0015 | | -0.1 | | | -0.1 | | | -0.1 |
| $x^2$ | | | | -0.23 | -0.27 | 1.8 | | | | | | |
| $y^2$ | | | | 0.023 | -0.022 | 0.081 | | | | | | |
| $z^2$ | | | | 0.00016 | -0.00031 | | | | | | | |
| $xy$ | | | | 0.063 | 0.22 | -0.8 | | | | | | |
| $xz$ | | | | -0.063 | -0.074 | | | | | | | |
| $yz$ | | | | 0.0019 | 0.0089 | | | | | | | |
| $h(x)$ | | | | | | | | | | | | |
| $h(y)$ | | | | 0.0063 | | | | | | | | |
| $h(z)$ | 1 | | | 0.51 | | | 0.26 | | | 0.93 | | |

**b**




Green = true positive; orange = false positive; rose = missed term. Numbers show coefficients.

Dots show noisy test observations; colored lines show selected-model rollouts.

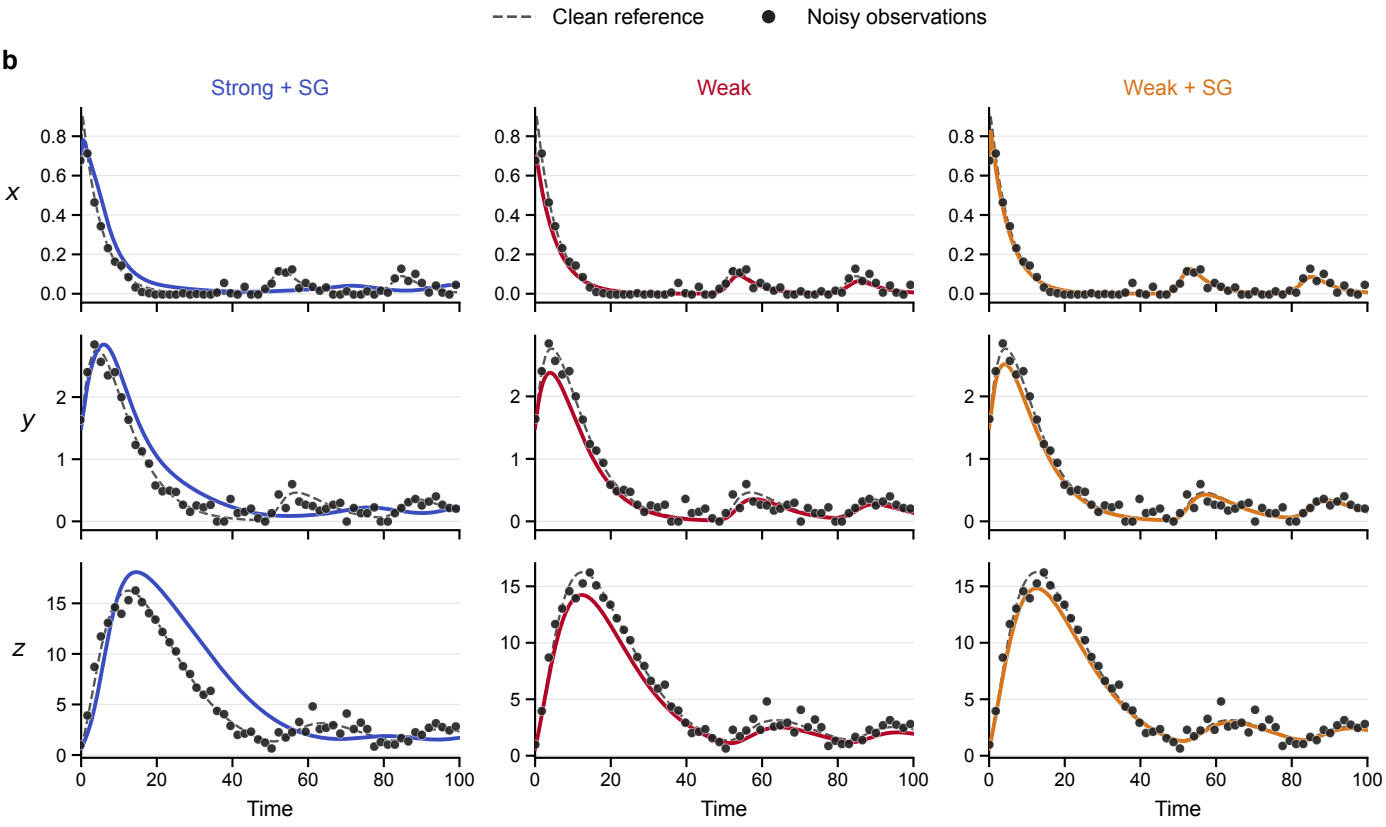
## Figure S1 | 15% noise, seed 11

Selected coefficient matrices and first held-out trajectory

**a**

| | Truth | | | Strong + SG | | | Weak | | | Weak + SG | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ |
| $x$ | -0.2 | 1 | | 0.15 | 0.93 | | -0.19 | 1.1 | | -0.2 | 0.95 | |
| $y$ | | -0.15 | 1 | -0.05 | -0.11 | 0.98 | | -0.16 | 1 | | -0.15 | 0.98 |
| $z$ | | | -0.1 | | -0.0034 | -0.11 | | | -0.098 | | | -0.096 |
| $x^2$ | | | | -0.19 | 0.36 | 3.7 | | | | | | |
| $y^2$ | | | | 0.045 | 0.043 | 0.47 | | | | | | |
| $z^2$ | | | | 0.00013 | 0.00064 | 0.0048 | | | | | | |
| $xy$ | | | | -0.042 | -0.31 | -3.4 | | | | | | |
| $xz$ | | | | -0.078 | 0.043 | 0.55 | | | | | | |
| $yz$ | | | | 0.0011 | -0.012 | -0.099 | | | | | | |
| $h(x)$ | | | | | | | | | | | | |
| $h(y)$ | | | | 0.0048 | | | | | | | | |
| $h(z)$ | 1 | | | 0.25 | | | 0.14 | | | 0.7 | | |

**b**




Green = true positive; orange = false positive; rose = missed term. Numbers show coefficients.

Dots show noisy test observations; colored lines show selected-model rollouts.

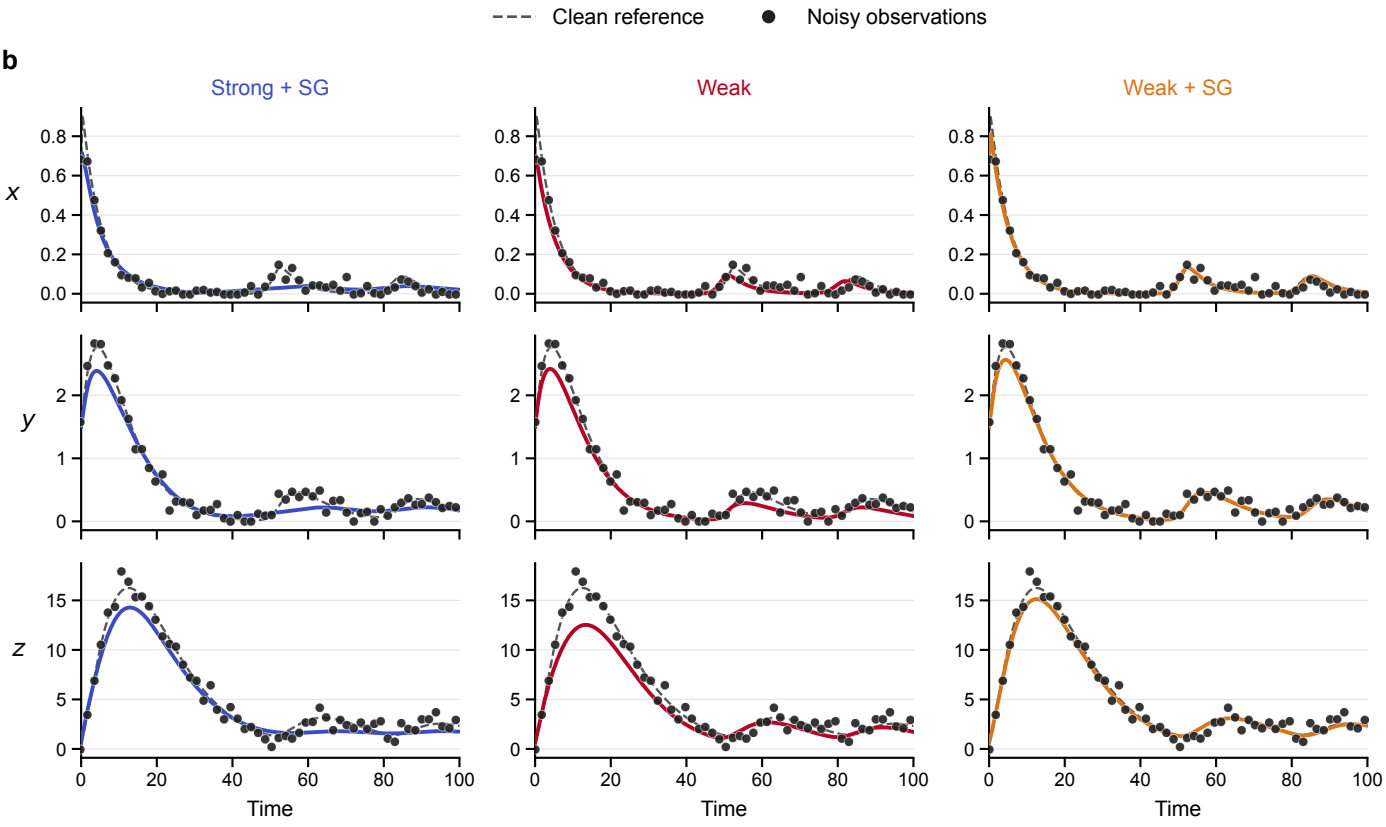
## Figure S1 | 15% noise, seed 22

Selected coefficient matrices and first held-out trajectory

**a**

| | Truth | | | Strong + SG | | | Weak | | | Weak + SG | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ |
| $x$ | -0.2 | 1 | | 0.16 | 0.95 | 1.9 | -0.19 | 0.78 | | -0.19 | 0.95 | |
| $y$ | | -0.15 | 1 | -0.055 | -0.15 | 0.63 | | -0.13 | 1.7 | | -0.14 | 1 |
| $z$ | | | -0.1 | -0.0008 | | -0.055 | | | -0.13 | | | -0.1 |
| $x^2$ | | | | -0.26 | | -1.3 | | 0.42 | | | | |
| $y^2$ | | | | 0.04 | -0.035 | 0.012 | | | -0.34 | | | |
| $z^2$ | | | | 0.00015 | -0.00027 | -0.0057 | | | | | | |
| $xy$ | | | | -0.0055 | 0.077 | 0.71 | | | | | | |
| $xz$ | | | | -0.081 | 0.016 | -0.66 | | | | | | |
| $yz$ | | | | 0.0027 | 0.0049 | 0.076 | | | | | | |
| $h(x)$ | | | | | | | | | | | | |
| $h(y)$ | | | | 0.0072 | | -0.066 | | | | | | |
| $h(z)$ | 1 | | | 0.18 | -0.12 | 0.4 | 0.14 | | | 0.65 | | |

**b**




Green = true positive; orange = false positive; rose = missed term. Numbers show coefficients.

Dots show noisy test observations; colored lines show selected-model rollouts.

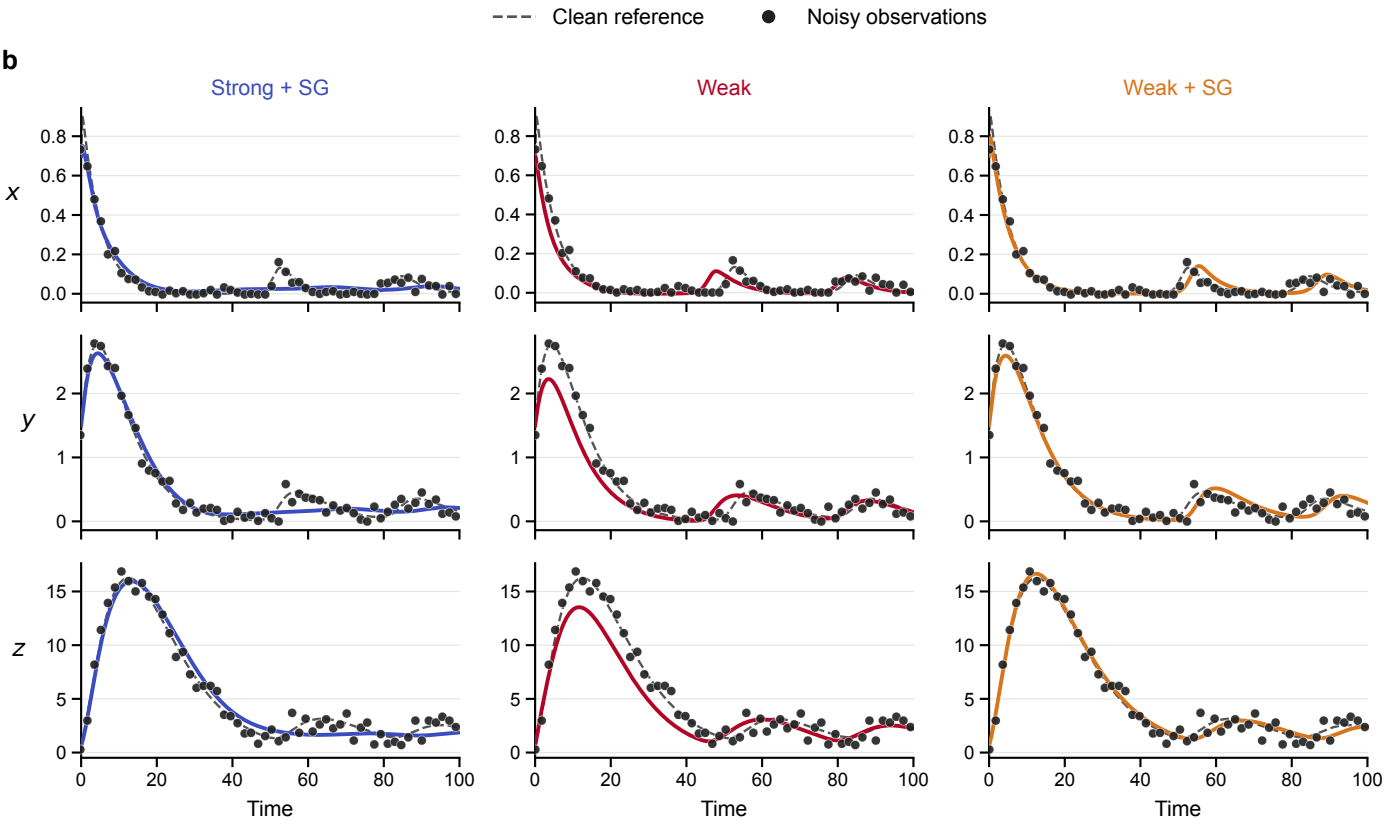
## Figure S1 | 15% noise, seed 33

Selected coefficient matrices and first held-out trajectory

**a**

| | Truth | | | Strong + SG | | | Weak | | | Weak + SG | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ |
| $x$ | -0.2 | 1 | | 0.18 | 0.98 | -0.85 | -0.11 | 0.87 | 0.76 | -0.19 | 1 | |
| $y$ | | -0.15 | 1 | -0.063 | -0.15 | 1.2 | -0.0075 | -0.1 | 1.1 | | -0.15 | 0.73 |
| $z$ | | | -0.1 | 0.001 | | -0.11 | | | -0.12 | | | -0.091 |
| $x^2$ | | | | -0.11 | | | | | | | | |
| $y^2$ | | | | 0.049 | -0.0093 | | 0.0058 | | | | | 0.14 |
| $z^2$ | | | | | -0.00018 | 0.0022 | | | | | | |
| $xy$ | | | | -0.11 | 0.073 | | -0.06 | | | | | |
| $xz$ | | | | -0.069 | -0.021 | 0.21 | | | -0.3 | | | |
| $yz$ | | | | 0.0021 | 0.0036 | -0.036 | | -0.0044 | 0.026 | | | |
| $h(x)$ | | | | | | | | | | | | |
| $h(y)$ | | | | 0.0044 | | | | | | | | |
| $h(z)$ | 1 | | | 0.26 | | | 0.11 | | | 0.53 | | |

**b**

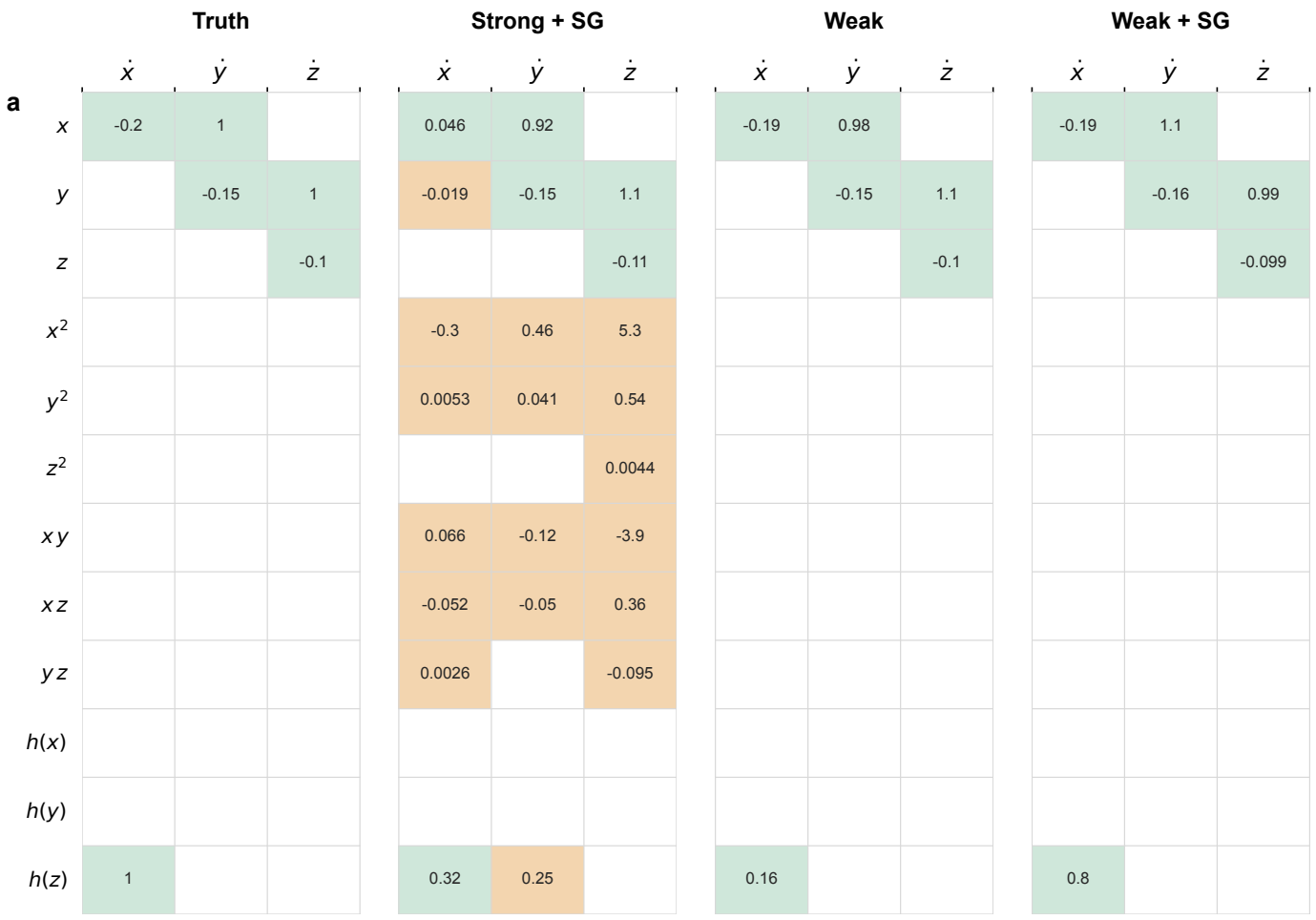


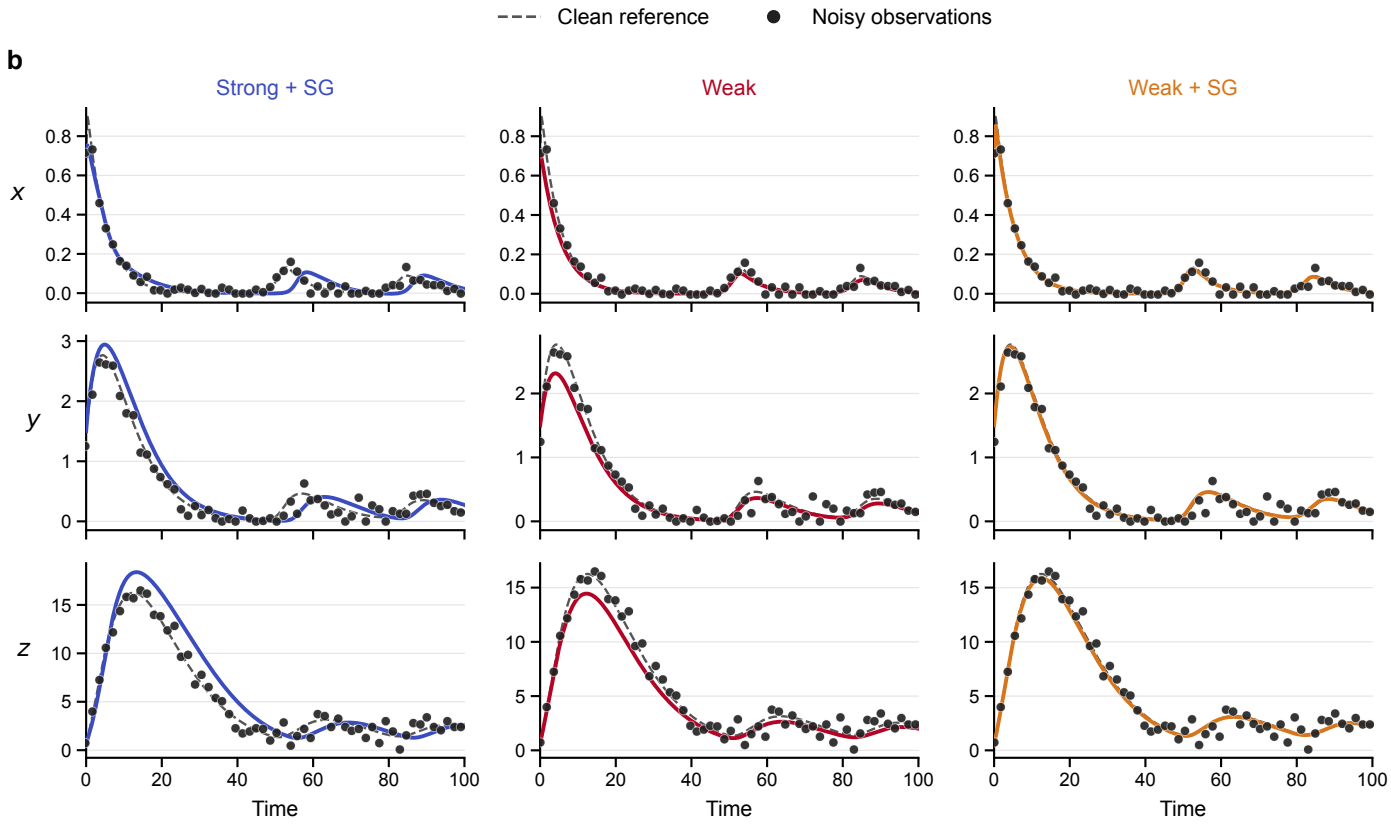

Green = true positive; orange = false positive; rose = missed term. Numbers show coefficients.

Dots show noisy test observations; colored lines show selected-model rollouts.

## Figure S1 | 15% noise, seed 44

Selected coefficient matrices and first held-out trajectory




Green = true positive; orange = false positive; rose = missed term. Numbers show coefficients.

Dots show noisy test observations; colored lines show selected-model rollouts.

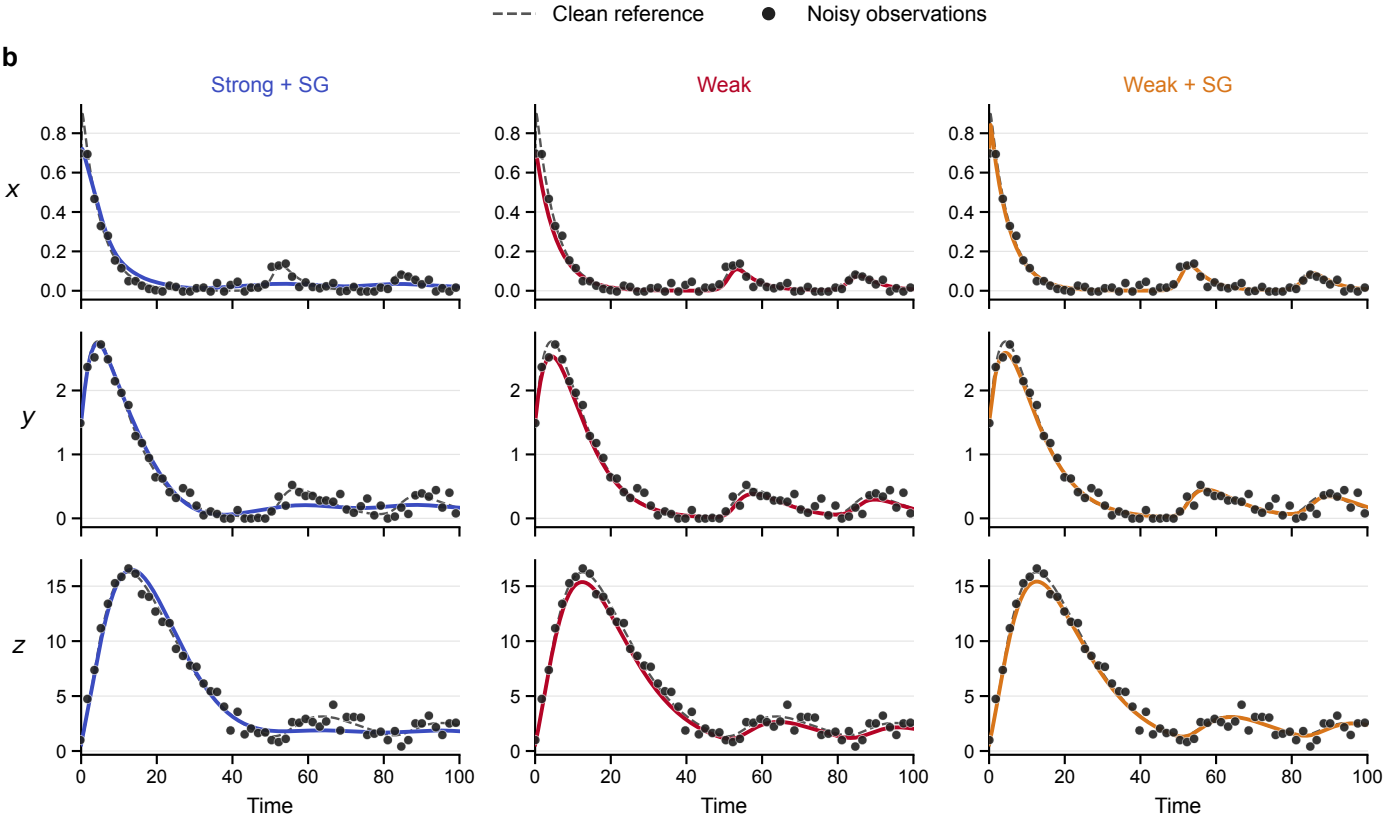
## Figure S1 | 15% noise, seed 55

Selected coefficient matrices and first held-out trajectory

**a**

| | Truth | | | Strong + SG | | | Weak | | | Weak + SG | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ |
| $x$ | -0.2 | 1 | | 0.21 | 1.1 | | -0.19 | 1 | | -0.2 | 0.93 | |
| $y$ | | -0.15 | 1 | -0.058 | -0.16 | 0.96 | | -0.15 | 1 | | -0.14 | 1 |
| $z$ | | | -0.1 | -0.0011 | | -0.1 | | | -0.1 | | | -0.1 |
| $x^2$ | | | | -0.4 | -0.4 | 2.7 | | | | | | |
| $y^2$ | | | | 0.04 | -0.036 | 0.13 | | 0.041 | | | | |
| $z^2$ | | | | 0.00021 | -0.00046 | | | | | | | |
| $xy$ | | | | 0.04 | 0.33 | -1.2 | | | | | | |
| $xz$ | | | | -0.091 | -0.1 | | | -0.079 | | | | |
| $yz$ | | | | 0.0025 | 0.013 | | | | | | | |
| $h(x)$ | | | | | | | | | | | | |
| $h(y)$ | | | | 0.0077 | | | | | | | | |
| $h(z)$ | 1 | | | 0.16 | | | 0.16 | | | 0.78 | | |

**b**




Green = true positive; orange = false positive; rose = missed term. Numbers show coefficients.

Dots show noisy test observations; colored lines show selected-model rollouts.

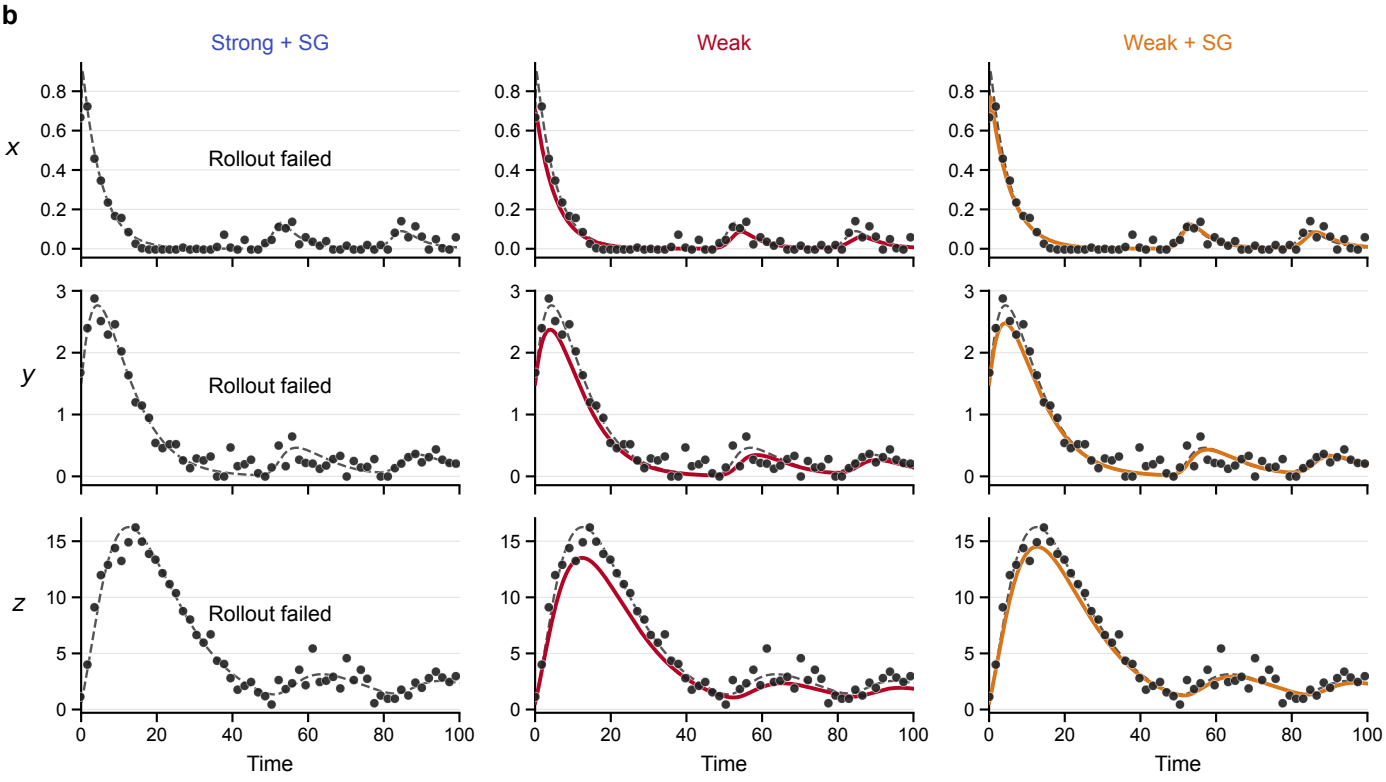
## Figure S1 | 20% noise, seed 11

Selected coefficient matrices and first held-out trajectory

**a**

| | Truth | | | Strong + SG | | | Weak | | | Weak + SG | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ |
| $x$ | -0.2 | 1 | | 0.24 | 0.89 | -0.55 | -0.19 | 1.1 | | -0.18 | 0.97 | |
| $y$ | | -0.15 | 1 | -0.062 | -0.1 | 1.1 | | -0.15 | 0.95 | | -0.15 | 0.98 |
| $z$ | | | -0.1 | 0.00024 | -0.0045 | -0.13 | | | -0.096 | | | -0.099 |
| $x^2$ | | | | -0.27 | 0.48 | 5.6 | | | | | | |
| $y^2$ | | | | 0.046 | 0.054 | 0.65 | | | | | | |
| $z^2$ | | | | 0.00014 | 0.00081 | 0.0077 | | | | | | |
| $xy$ | | | | -0.057 | -0.4 | -4.7 | | | | | | |
| $xz$ | | | | -0.075 | 0.058 | 0.75 | | | | | | |
| $yz$ | | | | 0.0014 | -0.015 | -0.14 | | | | | | |
| $h(x)$ | | | | | | | | | | | | |
| $h(y)$ | | | | 0.0052 | | 0.026 | | | | | | |
| $h(z)$ | 1 | | | 0.08 | | -0.53 | 0.093 | | | 0.46 | | |

**b**




Green = true positive; orange = false positive; rose = missed term. Numbers show coefficients.

Dots show noisy test observations; colored lines show selected-model rollouts.

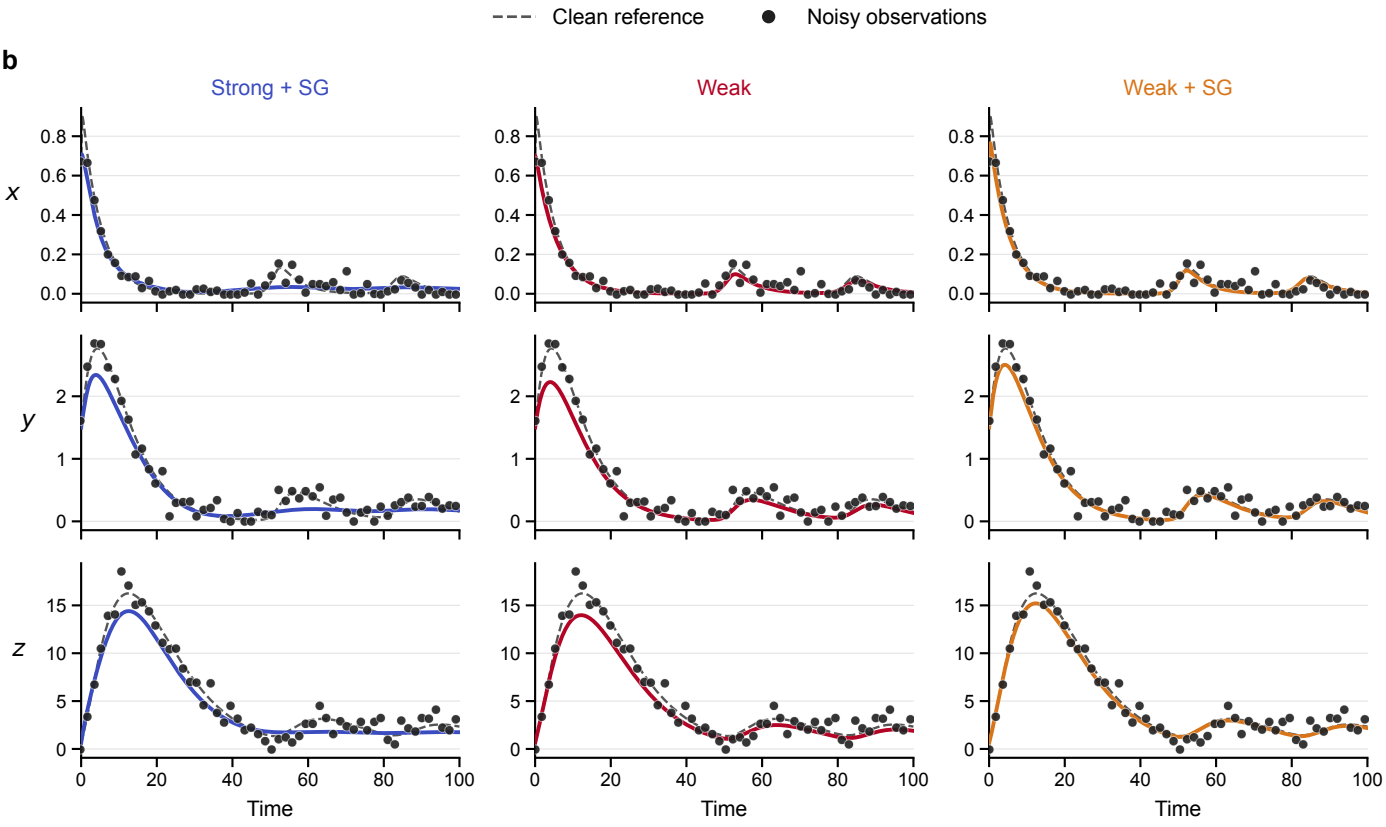
## Figure S1 | 20% noise, seed 22

Selected coefficient matrices and first held-out trajectory

**a**

| | Truth | | | Strong + SG | | | Weak | | | Weak + SG | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ |
| $x$ | -0.2 | 1 | | 0.21 | 0.86 | 2.3 | -0.18 | 0.93 | | -0.19 | 1 | |
| $y$ | | -0.15 | 1 | -0.061 | -0.13 | 0.51 | | -0.15 | 1.1 | | -0.15 | 1 |
| $z$ | | | -0.1 | -0.00098 | 0.0019 | -0.08 | | | -0.11 | | | -0.1 |
| $x^2$ | | | | -0.29 | | -1.2 | | | | | | |
| $y^2$ | | | | 0.046 | -0.069 | 0.16 | | | | | | |
| $z^2$ | | | | 0.00019 | -0.00064 | -0.0036 | | | | | | |
| $xy$ | | | | | 0.14 | 0.76 | | | | | | |
| $xz$ | | | | -0.095 | 0.041 | -0.92 | | | | | | |
| $yz$ | | | | 0.0029 | 0.0078 | 0.085 | | | | | | |
| $h(x)$ | | | | | | | | | | | | |
| $h(y)$ | | | | 0.0079 | -0.0054 | | | | | | | |
| $h(z)$ | 1 | | | 0.11 | -0.075 | 0.17 | 0.098 | | | 0.46 | | |

**b**




Green = true positive; orange = false positive; rose = missed term. Numbers show coefficients.

Dots show noisy test observations; colored lines show selected-model rollouts.

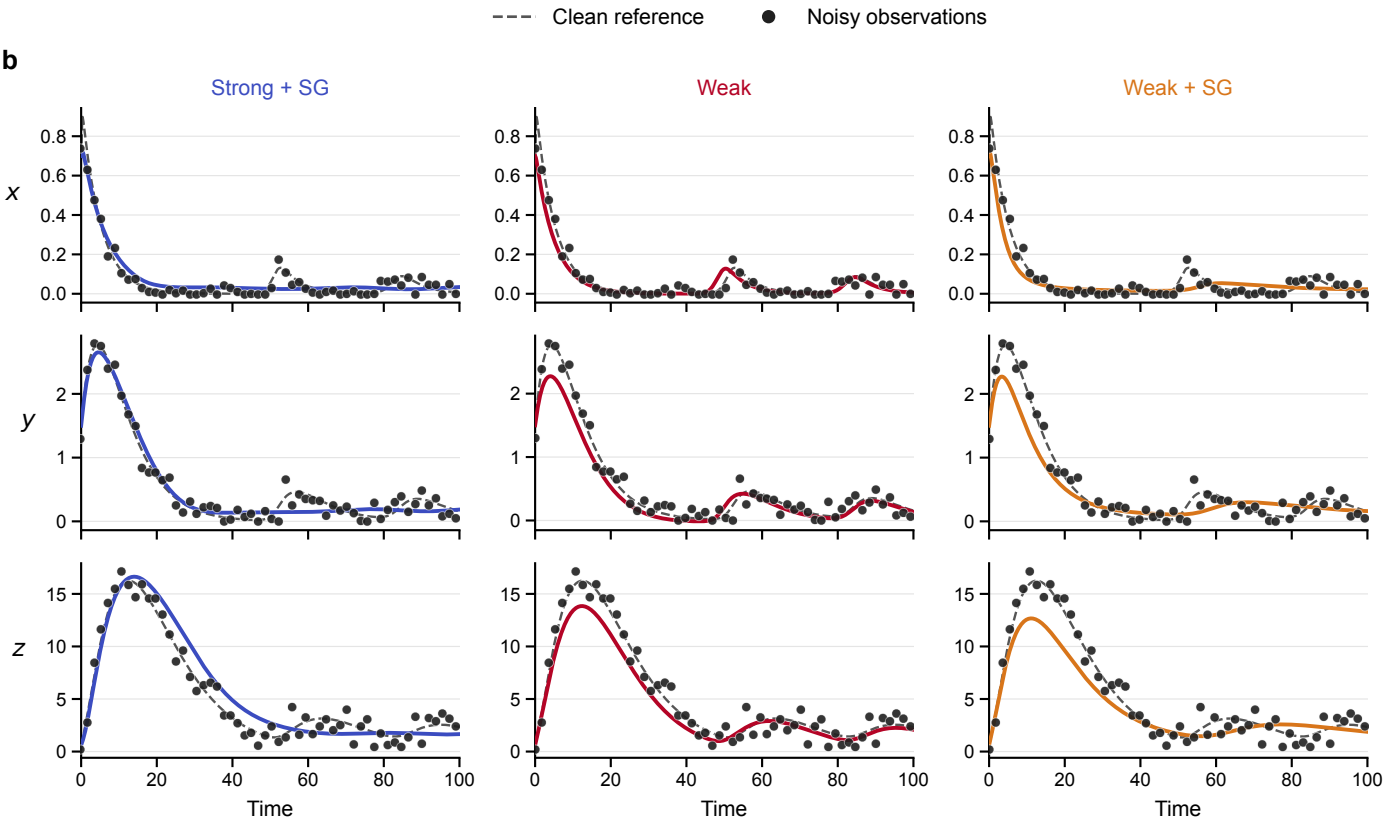
## Figure S1 | 20% noise, seed 33

Selected coefficient matrices and first held-out trajectory

**a**

| | Truth | | | Strong + SG | | | Weak | | | Weak + SG | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ |
| $x$ | -0.2 | 1 | | 0.21 | 0.88 | | -0.2 | 0.91 | | 0 | 1 | |
| $y$ | | -0.15 | 1 | -0.064 | -0.14 | 1.1 | | -0.13 | 1 | | -0.15 | 1.1 |
| $z$ | | | -0.1 | | | -0.11 | | -0.002 | -0.099 | | | -0.11 |
| $x^2$ | | | | -0.061 | 0.098 | -1.5 | | | | | | |
| $y^2$ | | | | 0.066 | -0.016 | 0.029 | | | | | | |
| $z^2$ | | | | 0.00023 | -0.00029 | 0.0033 | | | | | | |
| $xy$ | | | | -0.2 | 0.1 | | | | | -0.12 | | |
| $xz$ | | | | -0.063 | -0.022 | 0.25 | | | | | | |
| $yz$ | | | | -0.001 | 0.0048 | -0.043 | | | | | | |
| $h(x)$ | | | | | | | | | | | | |
| $h(y)$ | | | | 0.0059 | | | | | | | | |
| $h(z)$ | 1 | | | 0.15 | 0.057 | | 0.1 | | | 0.23 | | |

**b**




Green = true positive; orange = false positive; rose = missed term. Numbers show coefficients.

Dots show noisy test observations; colored lines show selected-model rollouts.

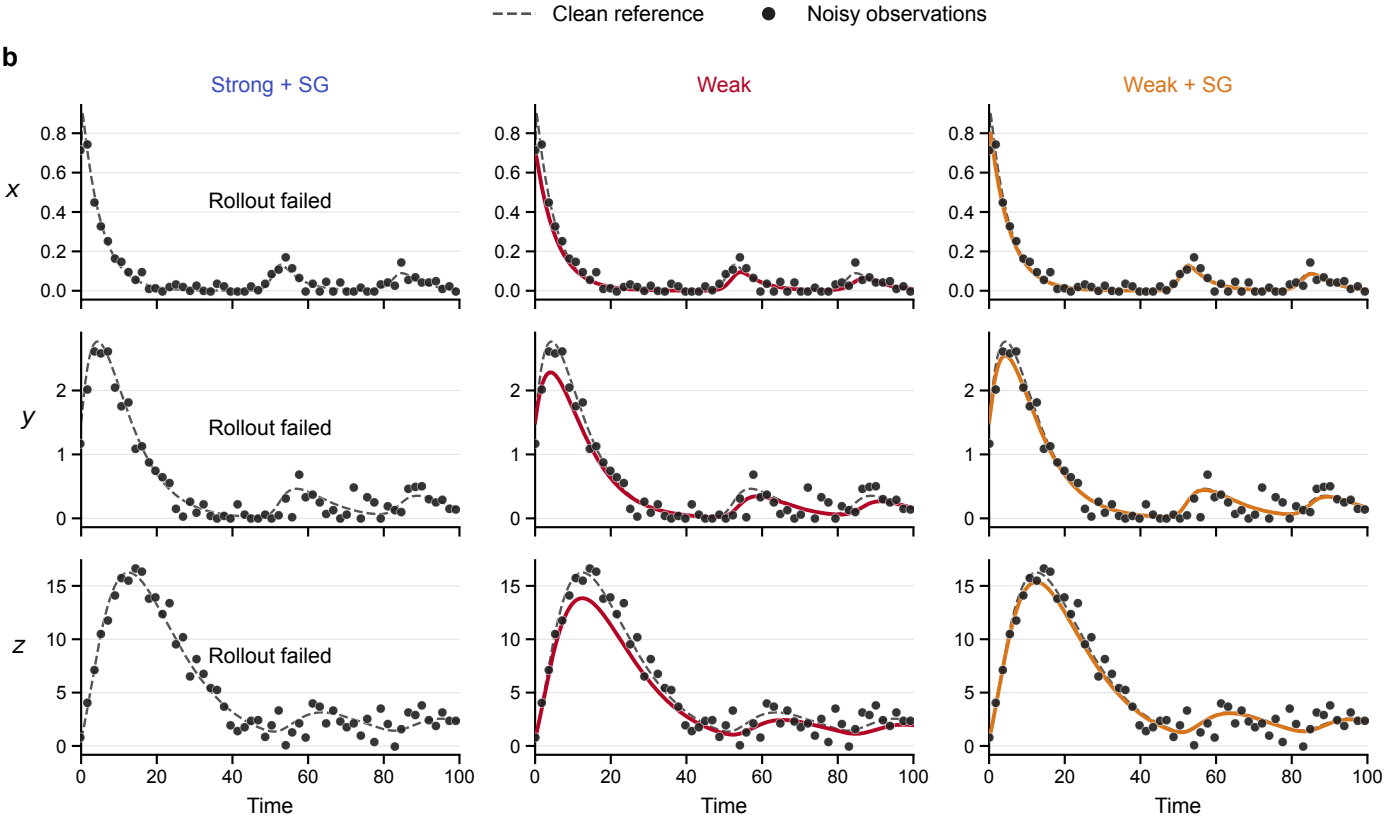
## Figure S1 | 20% noise, seed 44

Selected coefficient matrices and first held-out trajectory

**a**

| | Truth | | | Strong + SG | | | Weak | | | Weak + SG | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ |
| $x$ | -0.2 | 1 | | 0.17 | 0.82 | 1.5 | -0.18 | 0.95 | | -0.19 | 0.97 | |
| $y$ | | -0.15 | 1 | -0.046 | -0.11 | 0.88 | | -0.14 | 1 | | -0.15 | 1 |
| $z$ | | | -0.1 | -0.00057 | -0.0047 | -0.079 | | | -0.1 | | | -0.1 |
| $x^2$ | | | | -0.28 | 0.82 | 6.1 | | | | | | |
| $y^2$ | | | | 0.038 | 0.092 | 0.81 | | | | | | |
| $z^2$ | | | | 0.00024 | 0.00029 | 0.00055 | | | | | | |
| $xy$ | | | | -0.03 | -0.25 | -4.6 | | | | | | |
| $xz$ | | | | -0.065 | -0.08 | -0.024 | | | | | | |
| $yz$ | | | | 0.00016 | -0.0024 | -0.06 | | | | | | |
| $h(x)$ | | | | | -0.078 | -0.46 | | | | | | |
| $h(y)$ | | | | 0.0047 | 0.081 | 0.39 | | | | | | |
| $h(z)$ | 1 | | | 0.12 | 0.18 | -0.81 | 0.093 | | | 0.61 | | |

**b**




Green = true positive; orange = false positive; rose = missed term. Numbers show coefficients.

Dots show noisy test observations; colored lines show selected-model rollouts.

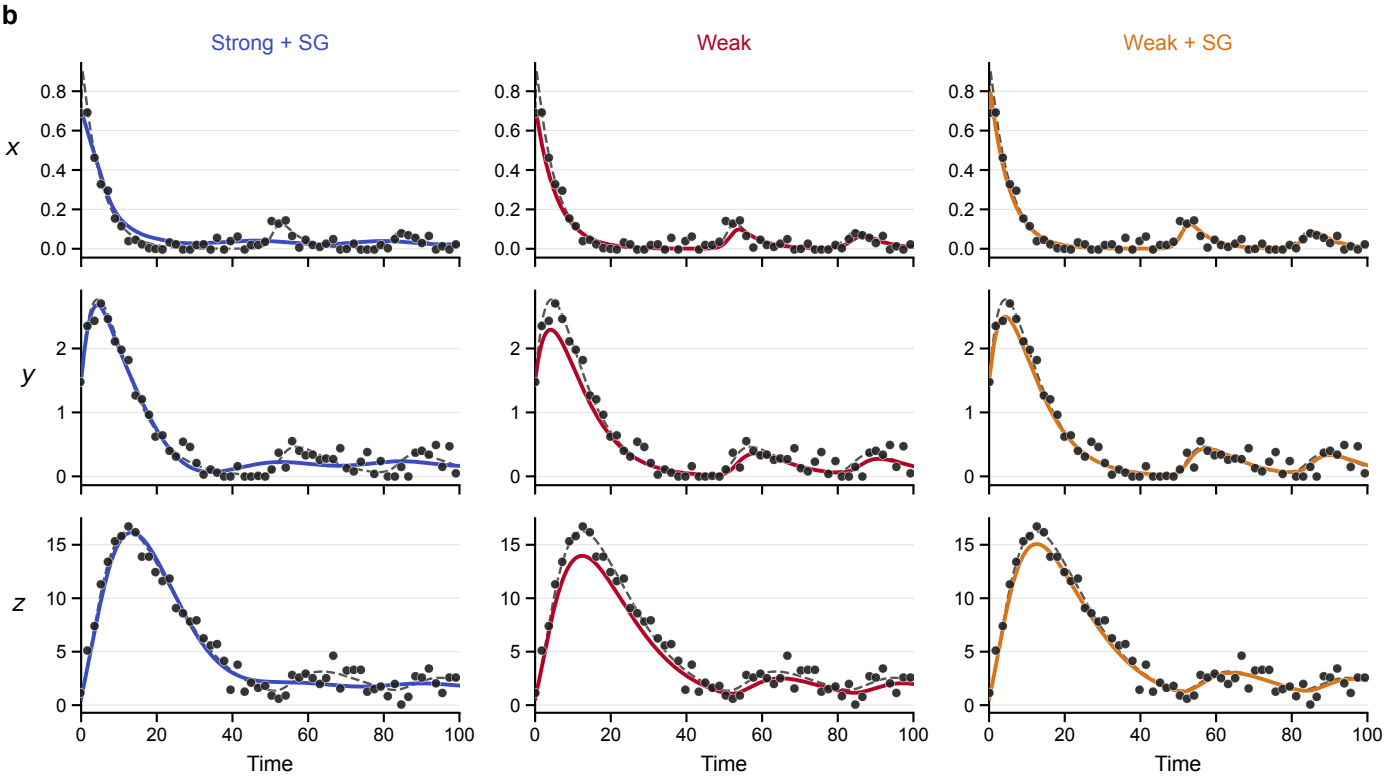
## Figure S1 | 20% noise, seed 55

Selected coefficient matrices and first held-out trajectory

**a**

| | Truth | | | Strong + SG | | | Weak | | | Weak + SG | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ | $\dot{x}$ | $\dot{y}$ | $\dot{z}$ |
| $x$ | -0.2 | 1 | | 0.28 | 1.2 | | -0.18 | 0.96 | | -0.19 | 0.95 | |
| $y$ | | -0.15 | 1 | -0.063 | -0.16 | 0.93 | | -0.15 | 1 | | -0.14 | 1 |
| $z$ | | | -0.1 | -0.0023 | | -0.1 | | | -0.1 | | | -0.1 |
| $x^2$ | | | | -0.38 | -0.49 | 3.6 | | | | | | |
| $y^2$ | | | | 0.054 | -0.047 | 0.17 | | | | | | |
| $z^2$ | | | | 0.00043 | -0.00056 | | | | | | | |
| $xy$ | | | | -0.044 | 0.42 | -1.6 | | | | | | |
| $xz$ | | | | -0.082 | -0.12 | | | | | | | |
| $yz$ | | | | | 0.016 | | | | | | | |
| $h(x)$ | | | | | | | | | | | | |
| $h(y)$ | | | | 0.01 | | | | | | | | |
| $h(z)$ | 1 | | | 0 | | | 0.11 | | | 0.56 | | |

**b**




Green = true positive; orange = false positive; rose = missed term. Numbers show coefficients.

Dots show noisy test observations; colored lines show selected-model rollouts.